

Regulatory Risk Coverage, Asset Prices, and Stress under the Sensitivities-Based Method of Capital Assessment

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September 2026

Abstract

Under the Fundamental Review of the Trading Book’s standardized approach, the capital a bank holds against its trading book is computed with a covariance matrix written by the regulator, over a factor list the regulator chooses. This paper studies what such a requirement does to equilibrium asset prices and how the distortions it creates behave in stress. Expected returns follow a two-beta CAPM, with one beta for aggregate risk and one for the regulator’s factor list. The wedge between regulatory and true risk splits into a calibration component that rescaling risk weights removes and a coverage component that no calibration can remove. Constrained intermediaries tilt toward uncovered risk, and a basis opens between otherwise identical exposures that differ only in regulatory coverage. The distortions are procyclical. In a calibration to the Basel parameters, impairing dealer capital and nonbank risk-bearing capacity together, as occurred in March 2020, widens the coverage basis by a factor of 29 and raises the largest pricing distortion from 43 to over 1,100 basis points. The indicators of regulatory tightness that supervisors conventionally watch do not show this, because the shadow cost of the requirement is hump-shaped in the rule’s stringency while the pricing distortions rise monotonically. A statistic assembled from the sensitivities banks already report, the *sensitivity gap*, rises with the distortion instead. A recalibration counterfactual and a welfare accounting measure the two components of the wedge.

Keywords: intermediary asset pricing; capital requirements; portfolio constraints in equilibrium; two-beta CAPM; regulatory risk measurement; FRTB; sensitivities-based method.

JEL codes: G12, G21, G28.

1 Introduction

Banks using the Standardized Approach of the Fundamental Review of the Trading Book face a regulation-defined covariance matrix to use when assessing the risk of their positions. This makes

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risk measurement a policy instrument. When a capital requirement binds, it is the risk model the regulator prescribes that determines which exposures are expensive for intermediaries to hold, and equilibrium prices adjust accordingly. Under the FRTB, the standardized approach was provided to serve as a backstop for banks too small to run internal models. Facing the profit-and-loss attribution test and the non-modelable risk factor framework, most large trading banks have declined to seek internal-model approval. Industry survey evidence indicates that institutions holding roughly 85% of trading-book capital under internal models plan internal-model coverage of only 15–40% under the new regime ([International Swaps and Derivatives Association and EY 2024](#)). The regime is arriving jurisdiction by jurisdiction. It is already live in Canada, Japan, and several other Basel Committee member countries. Following relief measures adopted in June 2026, it will go live in the European Union from January 2027. The United Kingdom plans to adopt in 2027–28. Recently the discussion has reopened in the United States with the timeline unresolved ([Basel Committee on Banking Supervision 2026](#); [European Commission 2026](#)). The landscape of bank regulatory regimes seems to be shifting toward a world where the marginal dealer in most of the world’s fixed income, equity, and derivatives markets uses a sensitivities-based method (SBM) of risk capital assessment. The SBM is a parametric portfolio risk model. Banks supply net sensitivities to a prescribed list of risk factors and the regulator supplies the risk weights, the correlations, and the list itself.

This paper asks two questions about such a rule. The first, and the paper’s principal concern, is what it does to equilibrium asset prices. The SBM’s risk model can differ from the risk that governs returns in two distinct ways. The first is calibration: its weights and correlations are statutory, so they adjust only slowly to market conditions. The second is coverage: the prescribed factor list is finite, and risks outside it, or netted away inside the sensitivity calculation, carry no charge however large they are. When the requirement binds, the difference between the regulator’s risk model and the true one becomes a portfolio-level pressure on returns. The second question is whether the officials who supervise the requirement can see any of this happening. The natural indicators of regulatory tightness (internal shadow prices on capital, asset-specific hurdle rates, reported headroom) all track the shadow cost of the constraint, and the paper shows that the shadow cost gives an incomplete picture of the risk borne by banks. The paper proposes an indicator, computed from data the SBM calculation already generates, that complements the measures of regulatory tightness supervisors typically use.

The paper answers these questions in a deliberately transparent setting, a static CAPM economy with two classes of investors. Traditional investors solve the usual mean–variance problem. Banks solve the same problem subject to a stylized capital requirement. Their riskless asset holdings must be at least as large as an SBM charge¹, computed as the square root of a quadratic form in the bank’s portfolio under a regulatory covariance matrix that the prescribed risk weights and correlations define. That matrix covers only the regulated factors, which are fewer than the factors that actually drive payoffs, so it is rank-deficient relative to the true covariance of returns. The

¹For expositional simplicity, this charge is modeled on the Delta component of the FRTB’s Sensitivities-Based Approach. Further work could add the vega and curvature portions of this calculation.

model thus allows the calibration and coverage components of the wedge to be studied separately. Given the nature of the model, prices can be solved for in equilibrium, and portfolios can be characterized.

Five sets of results follow. The first concerns prices. In equilibrium, every asset bears an extra required return in proportion to its marginal impact on bank balance sheets, meaning the funding a position requires plus the regulatory capital it consumes at the margin, at a uniform rate equal to the shadow cost of the requirement scaled by the banking sector’s share of the economy’s risk-bearing capacity. Expected returns obey a two-beta CAPM in which a regulatory beta, computed under the regulator’s covariance matrix rather than the true one, is priced alongside the market beta, and under a self-financed book even zero-beta assets earn a spread. The difference between the true covariance structure and the regulator’s imposed covariance leads to a regulatory wedge that creates cross-sectional distortions. This wedge can be decomposed into a calibration component the regulator could remove and a coverage component that no choice of weights can remove (see Proposition 1).

The second set of results concerns what parts of this equilibrium a supervisor can observe, and how the distortions behave in stress. The model shows that the shadow cost of the capital requirement is not monotone in the stringency of the rule. This is proven in a single-factor economy and demonstrated numerically in the calibrated multi-asset equilibrium. As risk weights scale up, the shadow cost of the requirement rises, peaks, and then falls, because banks shrink the charged book until a marginal dollar of capital buys little. The pricing distortions rise monotonically throughout, reaching roughly 200 basis points on equity exposure in the calibration even as measured tightness declines. A supervisor watching shadow prices might conclude that the requirement had stopped mattering where its effect on prices is largest. The model supplies an additional diagnostic measure, the *sensitivity gap*, the distance between the banking sector’s factor-level weighted sensitivities and their frictionless benchmark. The gap is proportional to the same equilibrium object that drives every pricing distortion, it rises monotonically through the region where the shadow cost falls, and its inputs are the weighted sensitivities of the SBM calculation itself, reported at each risk factor by every standardized-approach bank; the supplementary appendix shows how a supervisor assembles it from data the regime already collects. The distortions are also procyclical. In the calibrated stress scenarios, in which dealer equity and nonbank risk-bearing capacity are impaired together as they were in March 2020, the coverage basis widens by a factor of 29 and the largest pricing distortion rises from 43 to over 1,100 basis points.

The third set of results concerns quantities. The two sectors’ portfolios deviate from their frictionless benchmarks by opposite positions in a single tilt portfolio, whose regulatory part involves combinations of regulated-factor exposure. Two identically risky assets that differ only in the extent to which their risks are captured in the regulatory framework trade at different prices, a coverage basis for which the paper derives an explicit formula. Where unmeasured risk ends up depends on the financing margin. Under a pure capital requirement it gravitates to bank balance sheets as inventory, while under a self-financed book banks hold it only once its reward covers the resources

it ties up. The paper reports results at both ends of the financing spectrum, from a pure capital requirement to a fully self-financed book, and shows that the measured coverage basis depends on the financing assumption: under a pure capital requirement it is set by the charge on the on-list leg alone, while under a self-financed book it also reflects how much of the funding cost the off-list asset happens to bear.

The fourth set of results turns the wedge decomposition into a quantified policy experiment. The paper estimates a counterfactual recalibration where the covered factors' risk weights are set to be proportional to true factor volatilities, their prescribed correlations are set to the true correlations, and the overall scale is chosen to hold the charge on the frictionless bank book fixed. Measured tightness barely moves, but the composition distortions the statutory weights create are repaired. In this counterfactual, the capital charge on a dollar-matched hedged-equity book falls from 2.3 times its true risk to approximate parity, and in the stressed equilibria the cross-sectional spread of regulatory premia narrows by roughly a quarter. What remains is the coverage component. The uncharged relative-value exposure, and the banking sector's position in it, are untouched by any reweighting, because the factor list, not the weights, produces them.

The fifth result is a welfare accounting. The deadweight loss from the policy is calculated and its burden falls on issuers through the cost of capital, though under stress the constrained sector bears part of it as well. Section 5.3 puts magnitudes on these results using the standard's own risk weights: at a central tightness calibration the cross-sectional pricing hit is about 20 basis points per year on ten-year duration and 90–165 on single-name and emerging-market equity, the same order as documented post-crisis balance-sheet wedges, while the relative-value twins of those exposures earn under two basis points.

Three disclaimers are worth stating plainly. The model is static and uses only the delta portion of the SBM with a single correlation scenario. The coverage wedge is a cross-sectional object, arising from the rank of the regulatory covariance relative to the true one, that a dynamic economy would reproduce period by period, and the vega, curvature, default-risk, and scenario-max layers of the standard are left for other work. Nothing here speaks to the appropriate level of capital, only to the relative price and allocation structure a given level induces, a margin improvable without lowering aggregate capital. And no misreporting, ratings management, or evasion is assumed anywhere, so the distortions are properties of the rule, not of misconduct.

Section 2 describes related results in the literature. Section 3 sets up the economy and the charge. Section 4 derives the equilibrium. Section 5 characterizes the tilt toward unregulated risk, maps the results into FRTB language, calibrates the pricing hit on regulated exposures, constructs the sensitivity gap, and derives the welfare accounting. Section 6 solves the model at FRTB parameters and runs the recalibration counterfactual. Section 7 draws out the policy implications and concludes.

2 Related Literature

This paper connects four literatures: equilibrium asset pricing with portfolio constraints, models of risk-measure-based regulation, intermediary asset pricing and the empirics of post-crisis balance-sheet costs, and work on coarse regulatory risk measurement and the behavior it induces. The discussion moves from the model’s formal antecedents to its results, and closes with the small academic literature on the FRTB itself.

The model’s formal antecedents are the equilibrium models with portfolio constraints. Its machinery is related to [Black \(1972\)](#), whose restricted-borrowing CAPM also has the positive zero-beta intercept that reappears here as the funding wedge, the uniform rent every asset pays for space on a constrained balance sheet, in [Proposition 3](#). [Cuoco \(1997\)](#) gives a general theory of equilibrium with portfolio constraints in this class of economies, and [Basak and Croitoru \(2000\)](#) is the direct antecedent of the twin-asset result, showing that portfolio constraints allow two *redundant* securities to trade at different prices in equilibrium. The same law-of-one-price violation occurs in the limits-of-arbitrage literature, in which financially constrained arbitrageurs cannot fully close price gaps between identical or redundant assets ([Shleifer and Vishny 1997](#); [Gromb and Vayanos 2002](#)). The coverage basis of [Corollary 3](#) has a similar theoretical grounding, with the SBM charge in place of a collateral or equity constraint as the friction that creates the difference in equilibrium premia. Closest on the constraint’s functional form are the margin-CAPM of [Gârleanu and Pedersen \(2011\)](#) and the leverage-constrained investors of [Frazzini and Pedersen \(2014\)](#), in which each asset’s premium is proportional to its own margin or leverage cost. The SBM charge differs from a margin schedule in one essential respect. It is a joint quadratic form in the entire portfolio rather than a sum of per-asset terms, so the marginal capital consumed by any one asset depends on the whole book, hedges receive offsets, and the effective “margin” is endogenous to portfolio composition. This is what replaces the per-asset margin premium with a second *beta* ([Proposition 3](#)) and makes the constraint distort relative valuations, not just the level and slope of the security market line.

A second strand replaces the margin or leverage constraint with a statistical risk measure. It includes papers on optimal policies and prices under VaR constraints in general equilibrium ([Basak and Shapiro 2001](#); [Cuoco and Liu 2006](#)), the endogenous risk, procyclical leverage, and margin-liquidity spirals such constraints induce ([Danielsson et al. 2004](#); [Adrian and Shin 2014](#); [Brunnermeier and Pedersen 2009](#)), and, in the central banking literature, their general-equilibrium procyclicality and interaction with the risk-insensitive leverage ratio ([Covas and Fujita 2010](#); [Gambacorta and Karmakar 2018](#)). In all of these, the regulatory metric is computed under the *true* return distribution. The departure here is that the metric is computed under the regulator’s prescribed covariance matrix, which differs from the truth both in calibration and, because the regulated factors span only part of the true factor space, in rank. The results in this paper on the coverage and calibration wedge premia do not seem to have a direct counterpart in the VaR literature.

The tightness statistic through which the requirement enters prices connects the model to intermediary asset pricing. That intermediary capital is a priced state variable is the message of

He and Krishnamurthy (2013) theoretically and He et al. (2017) empirically. The shadow cost of bank capital, scaled by the banking sector’s share of aggregate risk-bearing capacity, is this paper’s regulation-specific counterpart, as discussed in Propositions 5 and 6. A large post-crisis empirical literature documents funding and regulatory balance-sheet costs in asset prices including covered interest parity deviations, funding value adjustments, balance-sheet “rental” costs, and negative swap spreads (Du et al. 2018; Andersen et al. 2019; Fleckenstein and Longstaff 2020; Jermann 2020; Klingler and Sundaresan 2019; Duffie 2018; Baker and Wurgler 2015). Kisin and Manela (2016) estimate the shadow cost of capital requirements directly from banks’ use of a costly loophole. These papers typically model the regulatory cost as a per-dollar or per-notional balance-sheet rent, uniform within an asset class. The contribution here is the cross-sectional structure of the rent implied by the SBM’s functional form, together with its equilibrium incidence across regulated and unregulated risks.

Closest to the results themselves is the literature on regulation with coarse or incomplete risk metrics and the behavior such metrics induce. Becker and Ivashina (2015) document that insurers subject to ratings-based capital requirements reach for yield *within* rating buckets, loading on the risk the metric does not distinguish. Boermans and van der Kroft (2024) find the same tilt in European institutions’ bond holdings, and Iannotta et al. (2019) develop the corresponding theory, and Opp et al. (2013) show how rating-contingent regulation distorts both the ratings themselves and the resulting allocation of capital. Miscalibration and strategic slack in model-based capital regulation (Behn et al. 2022; Plosser and Santos 2018), risk shifting under imperfectly observed capital (Dubecq et al. 2015), pre-crisis securitization that moved exposure outside the measured risk surface (Acharya et al. 2013), the price effects of regulatory pressure in the corporate bond market (Ellul et al. 2011), and the effect of statutory risk-based capital on insurers’ pricing (Kojen and Yogo 2015, 2022) all belong to the same family, as does the migration of activity to less-constrained balance sheets when capital is dear (Plantin 2015; Begenau and Landvoigt 2022). This paper differs in two ways. No misreporting, ratings management, or perimeter engineering is assumed: the tilt toward unmeasured risk (Proposition 7) is the optimum against a rank-deficient metric. And the general-equilibrium counterpart matters, since unmeasured risk is not merely relocated to constrained balance sheets but *repriced* for everyone (Corollary 3). Outside capital regulation, benchmarking and mandate frictions reprice covered relative to uncovered assets on the demand side. Basak and Pavlova (2013) show that institutions evaluated against an index bid up its constituents, and Kashyap et al. (2021) that a mandate referencing only a subset of assets subsidizes inclusion, the closest analog of the coverage basis priced here.

Finally, academic work on the FRTB itself is largely computational, for example Grajales and Medina Hurtado (2023) on SBM and expected-shortfall capital for options books and Avellone et al. (2025) on portfolios that minimize the internal-models charge, while industry studies document the migration of large banks onto the standardized approach (International Swaps and Derivatives Association and EY 2024), which is what makes the SBM the marginal capital metric in many markets. This paper takes the rule’s architecture as given and asks what it does to equilibrium

prices. One feature of that architecture is deliberately stripped out. That feature is the MAR21.6 requirement that the charge be computed under three correlation scenarios with the maximum binding. A maximum over correlation models is a maxmin objective, under which investors facing correlation ambiguity exhibit portfolio and price inertia (Illeditsch et al. 2021). The present model, with its single prescribed correlation matrix, is the smooth case to which the scenario-max layer adds a kinked, worst-case component.

3 Model

The economy is static, with two dates $t \in \{0, 1\}$. Trade occurs at date 0, and payoffs are realized and consumed at date 1. There is a riskless asset and N risky assets whose payoffs are driven by L underlying risk factors, of which the regulator’s capital framework recognizes only $K < L$. Two classes of investors trade. *Traditional investors* solve a standard mean–variance problem, and *constrained investors* (“banks”) face a stylized sensitivities-based capital requirement modeled on the delta component of the FRTB standardized approach.

3.1 Assets and risk factors

The riskless asset has an exogenous gross return $R_f \geq 1$ and is in perfectly elastic supply. The N risky assets have date-1 payoff vector $X \in \mathbb{R}^N$ generated by an L -factor structure,

$$X = \mu + Mf + \varepsilon, \quad (1)$$

where $\mu \in \mathbb{R}^N$ is the vector of expected payoffs, $f \in \mathbb{R}^L$ is a vector of risk factors with $f \sim \mathcal{N}(0, \Sigma_f)$, $\varepsilon \sim \mathcal{N}(0, \Sigma_\varepsilon)$ is idiosyncratic noise independent of f , and $M \in \mathbb{R}^{N \times L}$ is the matrix of factor exposures. Row i of M collects the *sensitivities* (deltas) of asset i ’s payoff to the L risk factors: $M_{i\ell} = \partial X_i / \partial f_\ell$. The payoff covariance matrix is

$$\Sigma := \text{Var}(X) = M\Sigma_f M' + \Sigma_\varepsilon, \quad (2)$$

and the following assumption is maintained throughout:

Assumption 1. $\Sigma_f \succ 0$, $\Sigma_\varepsilon \succeq 0$, and $\Sigma \succ 0$.

The regulatory framework recognizes only $K < L$ of these factors. Order the factors so that the regulated ones come first, and partition

$$f = \begin{pmatrix} f_r \\ f_u \end{pmatrix}, \quad M = [M_r \ M_u], \quad \Sigma_f = \begin{pmatrix} \Sigma_{rr} & \Sigma_{ru} \\ \Sigma_{ur} & \Sigma_{uu} \end{pmatrix}, \quad (3)$$

where $f_r \in \mathbb{R}^K$ collects the *regulated* factors, those for which the framework prescribes risk weights and correlations, and $f_u \in \mathbb{R}^{L-K}$ the *unregulated* factors, which are risks that are real, systematic, and priced by investors, but invisible to the capital calculation. $M_r \in \mathbb{R}^{N \times K}$ and $M_u \in \mathbb{R}^{N \times (L-K)}$ are the corresponding blocks of exposures.

Risky-asset prices $p \in \mathbb{R}^N$ are determined in equilibrium. A portfolio is a vector $\theta \in \mathbb{R}^N$ of share holdings together with a dollar position $b \in \mathbb{R}$ in the riskless asset. Short sales of risky assets and borrowing are permitted, subject only to the capital requirement introduced below. The *net regulated sensitivity* of portfolio θ is the vector

$$s(\theta) := M_r' \theta \in \mathbb{R}^K, \quad (4)$$

the model counterpart of the net sensitivities a bank reports to its supervisor for the risk factors the framework prescribes.

3.2 The regulatory capital charge

The regulator prescribes a risk weight $w_k > 0$ for each regulated factor $k = 1, \dots, K$ and a correlation matrix $\Gamma = (\gamma_{k\ell}) \in \mathbb{R}^{K \times K}$, symmetric and positive semidefinite with unit diagonal. Given a portfolio's net sensitivities $s = s(\theta)$, the *weighted sensitivities* are $WS_k = w_k s_k$, and the capital charge is the single-layer, delta-only analog of the SBM aggregation formula:²

$$\mathcal{K}(\theta) = \sqrt{\sum_{k=1}^K \sum_{\ell=1}^K \gamma_{k\ell} WS_k WS_\ell} = \sqrt{s(\theta)' \Omega s(\theta)}, \quad \Omega := D_w \Gamma D_w, \quad (5)$$

where $D_w = \text{diag}(w_1, \dots, w_K)$. Substituting $s(\theta) = M_r' \theta$ and defining the *regulatory covariance matrix*

$$\Sigma_{\mathcal{K}} := M_r \Omega M_r' \in \mathbb{R}^{N \times N}, \quad (6)$$

a positive semidefinite matrix of rank at most K , the charge takes the compact form

$$\mathcal{K}(\theta) = \sqrt{\theta' \Sigma_{\mathcal{K}} \theta}. \quad (7)$$

The function $\mathcal{K}$ is a seminorm on $\mathbb{R}^N$. It is nonnegative, positively homogeneous of degree one ($\mathcal{K}(t\theta) = |t| \mathcal{K}(\theta)$), and convex. It is differentiable everywhere except on the cone $\{\theta : \Sigma_{\mathcal{K}} \theta = 0\}$ of portfolios that carry no regulatory risk, with gradient

$$\nabla \mathcal{K}(\theta) = \frac{\Sigma_{\mathcal{K}} \theta}{\mathcal{K}(\theta)} \quad \text{wherever } \mathcal{K}(\theta) > 0. \quad (8)$$

Equation (7) says that the SBM charges capital as if portfolio risk were $\sqrt{\theta' \Sigma_{\mathcal{K}} \theta}$, that is, as if the covariance matrix of asset payoffs were $\Sigma_{\mathcal{K}}$ rather than the true Σ . Every parameter of $\Sigma_{\mathcal{K}}$ is set by the regulator. The bank supplies only its sensitivities. Furthermore, since the regulated factors span only part of the true factor space, the rank of $\Sigma_{\mathcal{K}}$ is capped at $K < L$, so whatever the risk weights, the regulator's matrix cannot reproduce true systematic risk unless the unregulated

²Relative to the full standardized approach (Basel Framework MAR21; [Basel Committee on Banking Supervision 2019](#)), the model makes three simplifications. First, it treats all K regulated factors as members of a single bucket, removing the two-layer structure in which within-bucket aggregation with correlations ρ is followed by cross-bucket aggregation with correlations γ . Second, it considers delta risk only, leaving out vega and curvature risk. Third, it works with a single prescribed Γ rather than the three correlation scenarios of MAR21.6. The scenario-max layer raises separate questions and is left for other work.

factors are irrelevant to payoffs. To keep track of both discrepancies at once, fix a benchmark scale $\phi > 0$ and define the *regulatory wedge*

$$\Delta := \Sigma_{\mathcal{K}} - \phi^2 (\Sigma - \Sigma_{\varepsilon}). \quad (9)$$

This wedge drives the pricing distortions in the model, and it has two sources: a *calibration wedge*, the difference between the prescribed weights and correlations and the true ones, and a *coverage wedge*, the difference between the true factor covariance and its projection onto the span of the regulated factors. Using $\Sigma - \Sigma_{\varepsilon} = M \Sigma_f M'$ and the partition (3), the wedge decomposes as

$$\Delta = \underbrace{M_r (\Omega - \phi^2 \Sigma_{rr}) M_r'}_{\text{calibration wedge}} - \underbrace{\phi^2 (M_r \Sigma_{ru} M_u' + M_u \Sigma_{ur} M_r' + M_u \Sigma_{uu} M_u')}_{\text{coverage wedge}}. \quad (10)$$

The calibration wedge vanishes under a *neutral calibration* of the covered factors. This happens when risk weights are proportional to the true volatilities of the regulated factors, $w_k = \phi \sigma_{r,k}$, and prescribed correlations equal to the true ones, $\Gamma = \text{corr}(f_r)$, so that $\Omega = \phi^2 \Sigma_{rr}$. The coverage wedge cannot be calibrated away, because it is the risk the framework does not capture. In FRTB terms, this risk is everything outside the prescribed risk-factor taxonomy, including non-modelable risk factors, residual basis risks, and risks the list simply omits.

The coverage wedge is a statement about the *rank* of the regulator's matrix that no choice of risk weights or correlations can escape. The next proposition bounds it below by the risk carried by the unregulated factors.

Proposition 1 (Coverage irreducibility). *Let $\Delta = \Sigma_{\mathcal{K}} - \phi^2 (\Sigma - \Sigma_{\varepsilon})$ be the wedge between the regulator's covariance matrix and the (benchmark-scaled) true systematic covariance of payoffs, as in (9), and let $M_u^{\perp}$ denote the part of the unregulated factors' payoff loadings that lies outside the span of the regulated factors. Then, however the regulator sets the risk weights and prescribed correlations,*

$$\|\Delta\| \geq \phi^2 \|M_u^{\perp} \Sigma_{uu} M_u^{\perp'}\| \geq 0, \quad (11)$$

where Σ_{uu} is the covariance of the unregulated factors and $\|\cdot\|$ is any measure of matrix size (formally, any unitarily invariant norm). The lower bound depends only on the risk carried by the uncovered exposure, not on the risk weights or correlations. The bound is strictly positive whenever the uncovered exposure is nonzero, so the regulatory and true risk models coincide (and the wedge can be driven to zero) only in the special case in which the regulated factors already span every priced source of risk.

Proposition 1 (proved in the supplementary appendix) is the formal counterpart of the paper's policy claim that the factor *list*, not the weights, is the binding design choice. Reweighting can drive the calibration wedge to zero, but the coverage wedge is bounded away from zero by an object the regulator does not control.

Unmeasured risk is also free. Because the charge depends on θ only through the regulated sensitivities $M_r' \theta$, any portfolio in the null space of M_r' carries a capital charge of zero. That null

space has dimension at least $N - K$ and contains not only pure idiosyncratic positions but every position whose risk runs through the unregulated factors f_u , so such a portfolio can carry arbitrarily large true, systematic risk while consuming no regulatory capital. Section 5.1 makes the resulting incentive precise in equilibrium.

3.3 Investors

There is a continuum of investors of total mass one, divided into two classes, *traditional investors* (superscript U , for unconstrained) of mass $m_U \in (0, 1)$, and *constrained investors* (superscript C) of mass $m_C = 1 - m_U$. Investors within a class are identical, and all are price takers.

An investor of class $j \in \{U, C\}$ enters date 0 with wealth W_0^j , chooses risky holdings $\theta \in \mathbb{R}^N$ and a riskless position $b \in \mathbb{R}$ subject to the budget constraint $b + p'\theta = W_0^j$, and consumes date-1 wealth

$$W_1 = R_f b + \theta' X = R_f W_0^j + \theta' (X - R_f p). \quad (12)$$

Preferences over terminal wealth are CARA with risk aversion $\eta_j > 0$. Given the Gaussian structure (1), expected utility maximization is equivalent to the mean-variance criterion

$$V^j(\theta) := \theta'(\mu - R_f p) - \frac{\eta_j}{2} \theta' \Sigma \theta. \quad (13)$$

Traditional investors. Traditional investors solve the unconstrained problem

$$\max_{\theta \in \mathbb{R}^N} V^U(\theta), \quad (14)$$

with the familiar solution

$$\theta^U = \frac{1}{\eta_U} \Sigma^{-1} (\mu - R_f p). \quad (15)$$

Constrained investors. Constrained investors face, in addition, a stylized capital requirement. Own resources must cover the capital charge on the risky book, together with a fraction $\psi \in [0, 1]$ of the *gross* value of that book,

$$W_0^C \geq \psi \sum_{i=1}^N p_i |\theta_i| + \mathcal{K}(\theta), \quad (16)$$

so that the constrained problem is

$$\max_{\theta \in \mathbb{R}^N} V^C(\theta) \quad \text{subject to} \quad \psi \sum_{i=1}^N p_i |\theta_i| + \sqrt{\theta' \Sigma \mathcal{K} \theta} \leq W_0^C. \quad (17)$$

The parameter ψ is equivalent to a leverage cap. Rearranging (16),

$$\sum_{i=1}^N p_i |\theta_i| \leq \frac{W_0^C - \mathcal{K}(\theta)}{\psi}, \quad (18)$$

so the gross value of the risky book may not exceed $1/\psi$ times the own resources that remain once the capital charge has been set aside. The constraint therefore caps the bank's gross leverage at $1/\psi$, and the two endpoints are the two polar caps. At $\psi = 1$ the cap is one, so the book must be funded entirely from own resources, the riskless position satisfies $b \geq \mathcal{K}(\theta)$, and the capital backing the requirement is held in riskless, liquid form. At $\psi = 0$ the cap is infinite and the requirement is a pure capital rule: positions may be debt-financed at the riskless rate and own resources are consumed only by the charge. This is the requirement as the standard states it, eligible capital being a liability-side quantity that does not move when a bank finances a purchase with debt. Intermediate values sit between these poles. A cap of about 33 times own resources, $\psi = 0.03$, is the Basel III Tier 1 leverage ratio applied on top of the risk charge, and tighter caps read as a haircut or margin schedule on gross positions.³ Results are stated for general ψ , and Section 6 reports both endpoints.

A second feature of (16) is that funding is levied on *gross* rather than net position value. Under a net term $\psi p'\theta$ a short position would *release* funding at full value, making short sales a source of regulatory capacity. That is not a property of any capital or collateral rule, since a short requires collateral posted to the securities lender and is counted gross under the leverage ratio. Section 6.4 shows that the choice is not innocuous for the paper's central prediction.

Requirement (16) is a deliberately stylized, static stand-in for the regulatory requirement that eligible capital be at least as large as the market risk charge, and ψ separates that requirement from the financing assumption attached to it. The distinction matters because the two enter prices differently. The charge $\mathcal{K}(\theta)$ is a function of portfolio composition and generates every cross-sectional result in the paper. The funding term is linear in gross value and generates a uniform level effect only. At $\psi = 1$, since $\mathcal{K} \geq 0$, the requirement implies $b \geq 0$, so the constrained investor can never finance risky positions by borrowing at the riskless rate and the constraint embeds a no-borrowing restriction on top of its risk-sensitive component; at $\psi = 0$ it does not. For every $\psi \in [0, 1]$ the constraint set is closed, convex, and positively homogeneous of degree one, and satisfies Slater's condition whenever $W_0^C > 0$ because $\theta = 0$ is strictly feasible, so (17) has a unique solution. The one cost of the gross formulation is differentiability: the funding term has a kink at $\theta_i = 0$, where the first-order condition holds with a subgradient. Alternative formulations (a charge levied against equity, a leverage bound on borrowing, a requirement against risk-weighted assets (Gambacorta and Karmakar 2018)) differ only in the funding component and are nested by the choice of ψ . In any formulation in which the shadow cost multiplies the marginal charge $\partial\mathcal{K}/\partial\theta_i$, the two-beta structure, the wedge premium, and the tilt toward unmeasured risk survive, because those results flow from the shape of $\mathcal{K}$ rather than from the financing assumption.

³The Basel framework imposes the leverage ratio and the risk-based requirement as two separate constraints, of which the tighter binds. Equation (16) adds the two instead, which keeps the constraint set convex with a single multiplier and lets ψ move continuously from one financing assumption to the other. The difference matters only when the leverage constraint alone would bind, a case the paper does not study.

3.4 First-order conditions and the distorted demand of banks

Let $\lambda \geq 0$ be the multiplier on the capital requirement in (17). At any solution with $\mathcal{K}(\theta) > 0$, the first-order condition is

$$\mu - R_f p = \eta_C \Sigma \theta^C + \lambda \psi D_\varsigma p + \lambda \frac{\Sigma_{\mathcal{K}} \theta^C}{\mathcal{K}(\theta^C)}, \quad \lambda \left[W_0^C - \psi \sum_i p_i |\theta_i^C| - \mathcal{K}(\theta^C) \right] = 0, \quad (19)$$

where $D_\varsigma := \text{diag}(\varsigma)$ collects the subgradients of the gross funding term, with $\varsigma_i = \text{sign}(\theta_i^C)$ wherever $\theta_i^C \neq 0$ and $\varsigma_i \in [-1, 1]$ determined in equilibrium wherever $\theta_i^C = 0$. Rearranging (19) isolates the two channels through which the requirement distorts demand:

$$\mu - R_f p - \lambda \psi D_\varsigma p = \left[\eta_C \Sigma + \frac{\lambda}{\mathcal{K}(\theta^C)} \Sigma_{\mathcal{K}} \right] \theta^C. \quad (20)$$

A binding capital requirement makes the constrained investor behave as if (i) her funding rate on asset i were $R_f + \lambda \psi \varsigma_i$ rather than R_f , a funding spread of $\lambda \psi$ levied on every dollar of own resources a position ties up, on long and short positions alike, and (ii) her risk aversion were augmented by an additional term $\lambda / \mathcal{K}(\theta^C)$ applied to the *regulator's* covariance matrix $\Sigma_{\mathcal{K}}$ rather than the true covariance Σ . Channel (i) is a level effect familiar from margin- and leverage-constrained models. Channel (ii) is the distinctive feature of a sensitivities-based regime, re-pricing risk *asset by asset* according to the regulator's risk weights and correlations. The impact of this additional risk term is the main object of study in this paper. If the regulator's risk calibration correctly weights the factor risks (the neutral calibration of Section 3.2), $\Sigma_{\mathcal{K}} \propto \Sigma - \Sigma_\varepsilon$ and channel (ii) collapses to a nearly uniform increase in effective risk aversion. If, however, market risks have drifted from the statutory risks given in $\Sigma_{\mathcal{K}}$, assets whose regulatory risk exceeds their true risk are penalized in the constrained investor's portfolio and must offer higher equilibrium returns to be held.

Lemma 1 in the next section states the resulting demand in closed form, Section 4.5 establishes the comparative statics of λ , and existence of equilibrium is established in the supplementary appendix.

3.5 Equilibrium

The risky assets are in fixed per-capita supply $\bar{\theta} \in \mathbb{R}_+^N$. The riskless asset is in perfectly elastic supply at the exogenous rate R_f .

Definition 1 (Equilibrium). An *equilibrium* is a price vector $p \in \mathbb{R}^N$ and portfolios (θ^U, θ^C) such that (i) θ^U solves (14) and θ^C solves (17) given p , and (ii) the market for every risky asset clears:

$$m_U \theta^U + m_C \theta^C = \bar{\theta}. \quad (21)$$

When the capital requirement is slack ($\lambda = 0$), (15) and (19) coincide up to risk aversion and the model reduces to the standard CAPM with harmonic-mean risk aversion $\bar{\eta} := (m_U / \eta_U + m_C / \eta_C)^{-1}$ and pricing equation $\mu - R_f p = \bar{\eta} \Sigma \bar{\theta}$. The analysis in Section 4 characterizes how prices depart from this benchmark when the requirement binds.

4 Equilibrium Characterization

This section derives the equilibrium in closed form. Proofs of all results are collected in the supplementary appendix. Throughout, $\pi := \mu - R_f p \in \mathbb{R}^N$ denotes the vector of risk premia in payoff space (expected payoff in excess of the payoff of p_i dollars invested at the riskless rate). The focus is on equilibria in which the capital requirement binds with $\mathcal{K}(\theta^C) > 0$. Corollary 1 below gives the condition on bank capital W_0^C that separates the binding and slack regimes.⁴

4.1 Individual demands

Lemma 1 (Optimal portfolios). *Fix a price vector p with strictly positive entries and suppose $W_0^C > 0$.*

- (i) *The traditional investor's problem (14) has the unique solution $\theta^U = \frac{1}{\eta_U} \Sigma^{-1} \pi$.*
- (ii) *The constrained problem (17) has a unique solution θ^C , and there exists a multiplier $\lambda \geq 0$ satisfying $\lambda[W_0^C - \psi \sum_i p_i |\theta_i^C| - \mathcal{K}(\theta^C)] = 0$ such that, whenever $\kappa := \mathcal{K}(\theta^C) > 0$,*

$$\theta^C = H(\lambda, \kappa)^{-1} [\mu - R_f p - \lambda \psi D_\varsigma p], \quad H(\lambda, \kappa) := \eta_C \Sigma + \frac{\lambda}{\kappa} \Sigma_\mathcal{K} \succ 0. \quad (22)$$

Equation (22) restates the two channels of Section 3.4 as a demand function. A binding requirement makes the bank discount asset i at the shadow-augmented rate $R_f + \lambda \psi \varsigma_i$ and price risk with the blended matrix $H(\lambda, \kappa)$, a combination of the true covariance matrix and the regulator's.

4.2 Market clearing and equilibrium prices

Market clearing requires that the two sectors together absorb the supply of every risky asset,

$$m_U \theta^U + m_C \theta^C = \bar{\theta}, \quad (23)$$

while the riskless market clears automatically through its elastic supply at R_f . Substituting the demands of Lemma 1 into (23) and rearranging yields the paper's central pricing equation.

Proposition 2 (Equilibrium pricing). *In any equilibrium in which the capital requirement binds with shadow cost $\lambda > 0$ and equilibrium charge $\kappa = \mathcal{K}(\theta^C) > 0$, risk premia satisfy*

$$\pi = \underbrace{\bar{\eta} \Sigma \bar{\theta}}_{\text{fundamental risk}} + \underbrace{\chi \lambda \psi D_\varsigma p}_{\text{funding wedge}} + \underbrace{\frac{\chi \lambda}{\kappa} \Sigma_\mathcal{K} \theta^C}_{\text{regulatory risk}}, \quad \chi := \frac{m_C \bar{\eta}}{\eta_C} = \frac{m_C / \eta_C}{m_U / \eta_U + m_C / \eta_C} \in (0, 1), \quad (24)$$

where $\bar{\eta} = (m_U / \eta_U + m_C / \eta_C)^{-1}$. Equivalently, whenever $\chi \lambda \psi < R_f$, equilibrium prices are

$$p_i = \frac{1}{R_f + \chi \lambda \psi \varsigma_i} \left[\mu - \bar{\eta} \Sigma \bar{\theta} - \frac{\chi \lambda}{\kappa} \Sigma_\mathcal{K} \theta^C \right]_i. \quad (25)$$

⁴Equilibria in which the requirement binds but $\mathcal{K}(\theta^C) = 0$ would require the constrained sector's entire book to lie in the zero-charge cone $\{\theta : \Sigma_\mathcal{K} \theta = 0\}$. This case is nongeneric and is set aside.

If instead the requirement is slack, then $\lambda = 0$ and (24)–(25) reduce to the standard CAPM: $\pi = \bar{\eta} \Sigma \bar{\theta}$ and $p = R_f^{-1}(\mu - \bar{\eta} \Sigma \bar{\theta})$.

Two features of (24) deserve comment.

First, the constraint enters prices only through the product $\chi\lambda$ and the gradient of the charge. Using $\nabla \mathcal{K}(\theta^C) = \Sigma_{\mathcal{K}} \theta^C / \kappa$ from (8), the pricing equation can be written compactly as

$$\pi = \bar{\eta} \Sigma \bar{\theta} + \chi \lambda \left[\psi D_{\varsigma} p + \nabla \mathcal{K}(\theta^C) \right]. \quad (26)$$

The bracketed vector is asset i 's *marginal balance-sheet impact*. One more share requires ψp_i dollars of own resources and consumes a further $\partial \mathcal{K} / \partial \theta_i$ dollars of regulatory capital. The funding component is signed by the constrained sector's position, so an asset it holds long pays $+\psi \chi \lambda$, one it holds short pays $-\psi \chi \lambda$, and one it holds at zero pays $\varsigma_i \psi \chi \lambda$ for an interior ς_i that market clearing determines. In equilibrium every asset bears an extra required return in proportion to that impact, at the uniform rate $\chi \lambda$, which is the shadow cost of the requirement, λ , scaled by the constrained sector's share χ of the economy's aggregate risk-bearing capacity. The scalar $\chi \lambda$ is thus the model's sufficient statistic for regulatory tightness, rising as bank capital falls, as risk weights rise, or as the constrained sector occupies a larger share of aggregate risk tolerance.

Second, fundamental risk and regulatory risk both have a role in equilibrium prices while the funding wedge scales prices proportionally. The first term of (24) compensates covariance with the market portfolio under the true covariance matrix Σ , and the third compensates covariance with the bank sector's book under the regulator's matrix $\Sigma_{\mathcal{K}}$. Assets whose regulatory covariance with the bank book is large relative to their true covariance with the market must pay investors for the capital they consume, not only for the risk they carry. On top of both, equation (25) discounts the expected payoff of every asset the constrained sector holds long at $R_f + \psi \chi \lambda > R_f$, so under a self-financed book a binding capital requirement anywhere in the economy raises the effective discount rate on every such asset, including assets that carry no regulatory charge at all. Under a pure capital requirement ($\psi = 0$) this level effect is absent.

4.3 The two-beta CAPM and the cross-section of expected returns

To express Proposition 2 in the language of returns, suppose equilibrium prices are strictly positive and let $R_i := X_i / p_i$ denote gross returns, $R_M := \bar{\theta}' X / p_M$ the market return with $p_M := p' \bar{\theta}$, and $R_C := \theta^C' X / p_C$ the return on the constrained sector's book with $p_C := p' \theta^C$, assumed positive. For any portfolio returns, write $\text{Cov}_{\mathcal{K}}(\cdot, \cdot)$ and $\text{Var}_{\mathcal{K}}(\cdot)$ for the covariance and variance computed under the regulator's matrix $\Sigma_{\mathcal{K}}$ in place of Σ (so $\text{Cov}_{\mathcal{K}}(R_i, R_C) = e_i' \Sigma_{\mathcal{K}} \theta^C / (p_i p_C)$, the covariance the regulator's risk model *assigns*, whether or not it coincides with the true covariance).

Proposition 3 (Two-beta CAPM). *Under the conditions of Proposition 2, expected returns satisfy*

$$\mathbb{E}[R_i] - R_f = \psi \chi \lambda \varsigma_i + \beta_i^M \pi_M + \beta_i^K \pi_K, \quad i = 1, \dots, N, \quad (27)$$

where

$$\beta_i^M := \frac{\text{Cov}(R_i, R_M)}{\text{Var}(R_M)}, \quad \beta_i^K := \frac{\text{Cov}_K(R_i, R_C)}{\text{Var}_K(R_C)}, \quad \pi_M := \bar{\eta} p_M \text{Var}(R_M), \quad \pi_K := \frac{\chi \lambda \kappa}{p_C}.$$

Expected excess returns are spanned by two betas and an intercept. The *regulatory beta* β_i^K measures comovement with the bank sector's book as measured by the regulator's risk model, and its price π_K is the tightness statistic $\chi \lambda$ scaled by the regulatory riskiness of the bank book per dollar of value. The intercept $\psi \chi \lambda \varsigma_i$ is a zero-beta spread in the sense of Black (1972). This intercept is common across every asset the constrained sector holds long. An asset with both betas zero then earns $R_f + \psi \chi \lambda$ because holding it requires balance sheet resources. Under a pure capital requirement ($\psi = 0$) the intercept vanishes and the security market line passes through R_f , so the entire effect of the rule is carried by the regulatory beta. All three departures from the frictionless CAPM are proportional to $\chi \lambda$ and vanish together as the constraint slackens.

Corollary 1 (Binding threshold). *Let $p^0 := R_f^{-1}(\mu - \bar{\eta} \Sigma \bar{\theta})$ denote frictionless CAPM prices. An equilibrium with slack requirement exists if and only if*

$$W_0^C \geq \frac{\bar{\eta}}{\eta_C} [\psi p^{0'} \bar{\theta} + \mathcal{K}(\bar{\theta})], \quad (28)$$

in which case it is the CAPM equilibrium of Proposition 2 with $\lambda = 0$ and $\theta^C = (\bar{\eta}/\eta_C) \bar{\theta}$. If (28) fails, the requirement binds ($\lambda > 0$) in every equilibrium.

The threshold has a transparent reading. In the frictionless CAPM the bank sector holds the fraction $\bar{\eta}/\eta_C$ of the market portfolio, and (28) asks whether bank equity covers the capital charge of that book plus whatever share ψ of its value must be self-financed. The frictionless book is long in every asset, so gross and net funding coincide at the threshold. Under a pure capital requirement the funding term drops out and the threshold is the charge on the frictionless book alone.

The regulatory term in (24) mixes compensation for systematic risk that the regulator measures with compensation for the wedge between regulatory and true risk. Substituting $\Sigma_K = \phi^2(\Sigma - \Sigma_\varepsilon) + \Delta$, with Δ the wedge of (9), into (27) splits the regulatory premium of asset i into

$$\beta_i^K \pi_K = \underbrace{\frac{\chi \lambda \phi^2}{\kappa} \frac{[(\Sigma - \Sigma_\varepsilon) \theta^C]_i}{p_i}}_{\text{systematic repricing}} + \underbrace{\frac{\chi \lambda}{\kappa} \frac{(\Delta \theta^C)_i}{p_i}}_{\text{wedge premium}}. \quad (29)$$

The first term repackages true systematic risk at a higher price and changes only the level of risk compensation. The second is the cross-sectional signature of the wedge. An asset whose risk the covered-factor calibration overstates, and which banks hold long, carries a positive wedge premium, and its price is depressed relative to fundamentals. The coverage wedge enters Δ with a negative sign, so exposures that run through the unregulated factors command less expected return than their true risk warrants.

To understand the pricing implications of the policy choice, consider two extremes. At one extreme, the regulator's matrix is right. If the regulated factors span all payoff risk and are

neutrally calibrated, so that $\Sigma_K = \phi^2 \Sigma$ (a counterfactual benchmark requiring $M_u = 0$, $\Sigma_\varepsilon = 0$, and hence $K \geq N$), then expected returns obey an exact one-factor CAPM in true market beta, $\mathbb{E}[R_i] - R_f = b \psi \varsigma_i + a p_M \text{Cov}(R_i, R_M)$ with $a > \bar{\eta}$ and $b > 0$ given in closed form in the supplementary appendix, whose zero-beta intercept is strictly positive on every asset the bank holds long under a self-financed book and zero under a pure capital requirement. A fully covering, perfectly calibrated sensitivities-based requirement raises required returns in proportion to the size of the position, as in [Black \(1972\)](#) and [Frazzini and Pedersen \(2014\)](#), shifting and tilting the security market line but leaving relative valuations undistorted. The opposite extreme assumes that the regulator's model omits some important risk factors entirely.

Corollary 2 (Zero-charge portfolios). *Let $z \in \mathbb{R}^N$ satisfy $M'_T z = 0$ and $p'z \neq 0$, and let $\varsigma_z := \sum_i \varsigma_i p_i z_i / p'z$ denote the value-weighted average of the funding subgradients over the positions in z , equal to one when the constrained sector is long every asset that z holds. Then the portfolio z has $\beta_z^K = 0$ and*

$$\mathbb{E}[R_z] - R_f = \psi \chi \lambda \varsigma_z + \beta_z^M \pi_M.$$

Zero-charge portfolios earn no regulatory premium, priced by fundamental beta plus the funding wedge alone, regardless of how much true risk they carry, idiosyncratic or systematic through the unregulated factors. Under a pure capital requirement they are priced by fundamental beta alone, exactly as in the frictionless economy.

Such positions are the cheapest risk on a constrained balance sheet, priced as if the requirement did not exist except for the funding wedge. Everything in between the two extremes is governed by (29). Section 5.1 develops the portfolio-composition counterpart of these results.

4.4 Equilibrium computation

Proposition 2 characterizes prices given the equilibrium values of $(\lambda, \kappa, \theta^C)$, which are themselves endogenous. Equilibrium therefore requires a fixed point. This fixed point has a convenient structure. For *given* (λ, κ) the model is linear, so the entire equilibrium reduces to two scalar consistency conditions.

Proposition 4 (Scalar reduction). *Fix a sign configuration $\varsigma \in \{-1, 0, +1\}^N$ for the constrained sector's book, where $\varsigma_i = 0$ means that $\theta_i^C = 0$ and the funding subgradient on asset i is left free. For each $(\lambda, \kappa) \in [0, \infty) \times (0, \infty)$, the system consisting of the demands of Lemma 1, the kink conditions, and market clearing (23) is linear, and it has a unique solution $(p(\lambda, \kappa), \theta^U(\lambda, \kappa), \theta^C(\lambda, \kappa))$, given in closed form in the supplementary appendix, for every (λ, κ) when the funding gradient is uniform across assets ($\psi = 0$, or $\varsigma = \mathbf{1}$) and for all (λ, κ) outside a closed set of measure zero otherwise. A price system with binding requirement is an equilibrium if and only if, for some configuration ς , (λ, κ) solves the two scalar equations*

$$\mathcal{K}(\theta^C(\lambda, \kappa)) = \kappa, \quad \psi \sum_{i=1}^N p_i(\lambda, \kappa) |\theta_i^C(\lambda, \kappa)| + \kappa = W_0^C, \quad (30)$$

and the solution is consistent with ς , meaning $\theta_i^C \varsigma_i > 0$ where $\varsigma_i \neq 0$ and the implied subgradient lies in $[-1, 1]$ where $\varsigma_i = 0$.

The first equation makes the conjectured charge consistent with the implied book, and the second is the binding requirement itself. Computing an equilibrium is therefore a two-dimensional root-finding problem for each of at most 3^N sign configurations, regardless of K , and a single such problem under a pure capital requirement. This is the basis of the calibrated illustration in Section 6.

An equilibrium exists for every level of bank equity, every configuration of risk weights, and every value of the financing parameter, under one mild condition on expected payoffs that is needed only under gross funding and holds comfortably in the calibration. The statement, Theorem S.1, and its proof are in the supplementary appendix. The argument turns on a feature of the CARA-normal structure: because traditional investors are unconstrained, market clearing expresses risk premia as an affine function of the bank sector's book alone, so an equilibrium is a fixed point of the bank's best response to the prices its own book induces, and that book is bounded a priori by fundamentals and preferences however the regulator sets (w, Γ) . Uniqueness is not claimed. In the single-factor economy of Proposition 6 the equilibrium is unique, and the numerical probe of Section 6 finds no equilibrium other than the reported one.

4.5 Comparative statics

Two results describe how the equilibrium moves with the model's policy-relevant parameters. The first is a partial-equilibrium result on the shadow cost of capital that holds in full generality. The second is a complete general-equilibrium characterization in the single-factor economy, in which the coverage wedge is shut down by construction and the statics isolate the funding and scale channels.

Proposition 5 (The shadow value of bank capital). *Fix strictly positive prices p and let $v(W)$ denote the value of the constrained problem (17) with equity $W > 0$. Then v is nondecreasing and concave, and any KKT multiplier $\lambda(W)$ is a supergradient of v at W :*

$$v(W') \leq v(W) + \lambda(W) (W' - W) \quad \text{for all } W' > 0.$$

Consequently multipliers are nonincreasing in equity, meaning $W < W'$ implies $\lambda(W) \geq \lambda(W')$, and $v'(W) = \lambda(W)$ wherever v is differentiable.

Holding prices fixed, the shadow cost of the requirement falls as bank capital rises, because λ is the marginal value of equity and the constraint is convex. The single-factor economy adds the price channels and delivers global, strict comparative statics.

Proposition 6 (The single-factor economy). *Let $N = 1$ with a single, fully covered risk factor ($K = L = 1$, $M = 1$), $\text{Var}(X) = \sigma^2 > 0$, risk weight $w > 0$, supply $\bar{\theta} > 0$, and bank equity $W := W_0^C > 0$, so that $\mathcal{K}(\theta) = w |\theta|$ and the threshold (28) reads $W \geq \bar{W} := (\bar{\eta}/\eta_C) (\psi p^0 + w) \bar{\theta}$ with $p^0 = R_f^{-1}(\mu - \bar{\eta}\sigma^2\bar{\theta})$. Suppose $W < \bar{W}$. Then:*

(i) (Uniqueness.) *There is a unique equilibrium. The price p^* is the unique root in $(-w/\psi, \infty)$, read as $(-\infty, \infty)$ at $\psi = 0$, of the market-clearing equation*

$$\frac{m_U}{\eta_U \sigma^2} (\mu - R_f p) + \frac{m_C W}{\psi p + w} = \bar{\theta}, \quad (31)$$

with $\theta^C = W/(\psi p^ + w) > 0$, $\theta^U = (\mu - R_f p^*)/(\eta_U \sigma^2) > 0$, and*

$$\lambda = \frac{\sigma^2}{\psi p^* + w} \left[\frac{\eta_U}{m_U} \bar{\theta} - \left(\frac{\eta_U m_C}{m_U} + \eta_C \right) \theta^C \right] > 0, \quad (32)$$

which is positive because $\theta^C < (\bar{\eta}/\eta_C) \bar{\theta}$. The bank holds less than its frictionless CAPM book.

(ii) (Bank equity.) *p^* and θ^C are strictly increasing in W . The risk premium $\mu - R_f p^*$ and the shadow cost λ are strictly decreasing in W .*

(iii) (Risk weights.) *The risk premium $\mu - R_f p^*$ is strictly increasing in w , with*

$$\lim_{w \rightarrow \infty} (\mu - R_f p^*) = \frac{\eta_U \sigma^2 \bar{\theta}}{m_U},$$

the premium of an economy in which the constrained sector holds no risky assets. The shadow cost is not monotone: $\lambda \rightarrow 0$ as $w \rightarrow \infty$, and if $\bar{w} := \eta_C W / (\bar{\eta} \bar{\theta}) - \psi p^0 > 0$ then also $\lambda \rightarrow 0$ as $w \downarrow \bar{w}$, so λ is hump-shaped on the binding region $(\bar{w}, \infty)$ while the premium distortion increases monotonically throughout. This holds for every $\psi \in [0, 1]$.

Part (iii) of Proposition 6 separates two objects that are easily conflated, tightness and the shadow cost. As risk weights rise, the *price* distortion, the gap between the equilibrium premium and its frictionless level $\bar{\eta} \sigma^2 \bar{\theta}$, rises monotonically, converging to full disintermediation of the banking sector. The *shadow cost* λ , by contrast, first rises and then falls back to zero, because with sufficiently high risk weights the bank holds almost nothing, so a marginal dollar of equity is worth little, even though prices are then maximally distorted. The right diagnostic in the model is quantities rather than shadow prices, namely the gap between the bank sector's book and its frictionless benchmark $(\bar{\eta}/\eta_C) \bar{\theta}$.

Section 6 explores the general case numerically. Under both a proportional change in risk weights and a change in bank capital, the single-factor comparative statics carry over to the calibrated multi-asset economy.

5 Asset Pricing Implications

5.1 The tilt toward unregulated risk

Proposition 2 characterizes prices. This section characterizes the quantities held by banks and how the capital requirement reallocates risk between the two sectors. It is shown that the entire reallocation can be characterized by a position in a single portfolio.

Proposition 7 (The tilt portfolio). *In any equilibrium in which the requirement binds ($\lambda > 0$, $\kappa = \mathcal{K}(\theta^C) > 0$), the two sectors' books deviate from their frictionless CAPM benchmarks by opposite positions in a single tilt portfolio τ :*

$$\theta^C = \frac{\bar{\eta}}{\eta_C} \bar{\theta} - \frac{(1-\chi)\lambda}{\eta_C} \tau, \quad \theta^U = \frac{\bar{\eta}}{\eta_U} \bar{\theta} + \frac{\chi\lambda}{\eta_U} \tau, \quad \tau := \Sigma^{-1} [\psi D_\varsigma p + \nabla \mathcal{K}(\theta^C)], \quad (33)$$

These two deviations from the benchmark holding $\bar{\theta}$ aggregate to zero. The tilt away from the benchmark can be decomposed into $\tau = \psi \tau^{\text{fund}} + \tau^{\text{reg}}$ with

$$\tau^{\text{fund}} := \Sigma^{-1} D_\varsigma p, \quad \tau^{\text{reg}} := \frac{1}{\kappa} \Sigma^{-1} M_r \Omega s(\theta^C) \in \Sigma^{-1} \text{col}(M_r),$$

so the regulatory component lies in the K -dimensional subspace $\Sigma^{-1} \text{col}(M_r)$. The constraint induces banks to shed combinations of regulated-factor exposure only. It never directly sheds exposure to the unregulated factors f_u .

The tilt is a mean-variance portfolio in balance-sheet cost. An unconstrained mean-variance investor holds $\eta^{-1} \Sigma^{-1} \pi$, the inverse covariance matrix applied to the vector of risk premia. The tilt portfolio has the same form with the reward vector π replaced by $\psi D_\varsigma p + \nabla \mathcal{K}(\theta^C)$, the per-asset marginal balance-sheet impact of the pricing equation (26). Where an ordinary investor tilts toward high expected return per unit of risk, the bank tilts *away* from high balance-sheet cost per unit of risk, shedding the portfolio that buys the most constraint relief for the least true variance. The pre-multiplication by Σ^{-1} nets correlated positions and gives credit for hedges, so the bank sheds the variance-efficient combination rather than each asset in proportion to its own charge. Substituting the equilibrium θ^U into the traditional investors' first-order condition recovers (26), in which the wedge and regulatory premia are the compensation traditional investors demand for holding the tilt, and because τ^{reg} is confined to a K -dimensional subspace built from the regulator's factor list, unregulated risk becomes the cheap input on a constrained balance sheet.

Equal risk aversion. With $\eta_U = \eta_C = \eta$, so that $\bar{\eta} = \eta$ and $\chi = m_C$, (33) collapses to

$$\theta^C = \bar{\theta} - \frac{m_U \lambda}{\eta} \tau, \quad \theta^U = \bar{\theta} + \frac{m_C \lambda}{\eta} \tau. \quad (34)$$

The benchmark is now the market portfolio itself, so every difference between the two books stems from the capital requirement, and that difference is characterized by a position in the portfolio τ . Each sector's deviation is scaled by the *other* sector's mass, which is what makes the two deviations aggregate to zero, and as $m_C \rightarrow 1$ the traditional investors' book diverges from the market portfolio without bound in the direction of τ , the quantity counterpart of the rising premia they charge for the service. The distance between the two books, $\theta^U - \theta^C = (\lambda/\eta) \tau$ is the shadow cost of capital measured in units of risk tolerance. This object motivates the sensitivity gap analysis in Section 5.4.

The sharpest expression of the tilt holds fundamentals fixed and varies only the regulatory label, comparing two assets that are statistically identical but for the factor list.

Corollary 3 (Twin assets, on-list versus off-list risk). *Let assets a and b have identical fundamentals (expected payoff μ_0 , payoff variance σ^2 , per-capita supply $\bar{\theta}_0 > 0$, mutually uncorrelated and uncorrelated with all other assets) and differ only in regulatory treatment. Asset a loads on a regulated factor with risk weight $w > 0$, uncorrelated under the regulator's correlation matrix Γ with every other regulated exposure in the economy, while b loads on an unregulated factor. Suppose the bank's equilibrium position in b is strictly positive. Then, in any binding equilibrium, $\pi_a > \pi_b$, equivalently $p_a = p_b - (\pi_a - \pi_b)/R_f < p_b$; the bank holds strictly less of the on-list twin ($\theta_a^C < \theta_b^C$) and traditional investors strictly more ($\theta_a^U > \theta_b^U$). If in addition the bank's position in a is strictly positive, the gap is*

$$\pi_a - \pi_b = \frac{m_C \sigma^2 \theta_b^C \frac{\zeta}{1+\zeta}}{\frac{m_U}{\eta_U} + \frac{m_C}{\eta_C} \frac{1+\lambda\psi/R_f}{1+\zeta}} > 0, \quad \zeta := \frac{\lambda w^2}{\kappa \eta_C \sigma^2}, \quad (35)$$

and, holding the economy-wide statistics $(\lambda, \kappa, \theta_b^C)$ fixed, the gap (35) is strictly increasing in the risk weight w and converges, as $w \rightarrow \infty$, to $\eta_U m_C \sigma^2 \theta_b^C / m_U$.

The previous result shows the coverage distortion in its simplest form, two assets with *identical* payoff distributions trading at strictly different prices because only one of them appears on the regulator's factor list. The gap (35) is a *coverage basis* that vanishes with λ , w , and m_C , widens with regulatory tightness, and predicts a clientele pattern with a testable signature. Off-list risk concentrates on bank balance sheets and earns less than its fundamental risk warrants, the on-list twin migrates to unconstrained investors and earns more, and the spread between them comoves with dealer capital tightness.

5.2 The model's predictions in FRTB language

The model's regulated factors f_r map to the delta risk factors that MAR21 prescribes for its seven risk classes (MAR21.8–21.14; [Basel Committee on Banking Supervision 2019](#)): yield curves sampled at ten tenor vertices for general interest rate risk, issuer credit spread curves for the credit spread risk classes, spot prices and repo rates for equity risk, and each exchange rate against the reporting currency for foreign exchange risk. The unregulated factors f_u are everything that moves trading-book values and is not on that list, or that is on it only after the netting embedded in the sensitivity calculation has removed it. The framework itself charges part of the residue separately: jump-to-default is charged under MAR22, and a *residual risk add-on* levies flat charges on gross notional for exotic underlyings and other residual risks (MAR23). That add-on is the regulation's own patch for the coverage wedge, and because it is notional-based rather than risk-based, *within* its coverage the marginal charge is again invariant to true risk.⁵

⁵The delta-only model abstracts from the vega and curvature charges and from the default risk capital requirement. The predictions below are therefore for delta-dominated, default-remote books (e.g. rates, FX, and linear equity exposure) and for relative comparisons within them.

Corollary 3 then predicts which exposures gravitate to standardized-approach banks, namely those whose *marginal* charge $\partial\mathcal{K}/\partial\theta_i$ is small relative to their true risk. Applying that criterion to the SBM’s netting and correlation structure points to several places where real risk goes largely uncharged.

1. *Same-factor relative-value positions carry no charge.* Two instruments mapping to the same risk factor net perfectly. Examples include the on-the-run/off-the-run Treasury basis, the cash–futures basis at a common vertex, and any specialness, liquidity, or deliverability premium between instruments the taxonomy treats as identical. These are pairs of assets that differ only in regulatory treatment, the case of Corollary 3. Such positions are likely to concentrate on standardized-approach bank balance sheets, where the spreads they earn are tighter than their true risk warrants and widen when dealer capital tightens.
2. *Cross-curve relative-value positions within a currency carry almost no capital charge.* Sensitivities to different curves in the same currency (e.g., OIS versus term-rate curves) and to different instruments of the same issuer (the bond–CDS basis) are aggregated at a prescribed correlation of 99.90% (MAR21.47 for different curves in the same currency, MAR21.54 for different curves of the same issuer). A \$1-for-\$1 ten-year cross-curve position carries $\sqrt{2(1 - 0.999)} \approx 1/22$ of the capital of an outright ten-year position, the same true basis risk at four percent of the measured risk. The model predicts dealer concentration in cross-curve and bond–CDS positions, with compressed spreads in calm times. This is the model’s rationalization of the balance-sheet-intensive relative-value trades documented in the intermediary literature of Section 2.
3. *Regulated portfolios will tilt away from high-risk-weight directional exposure.* At the other end of the taxonomy, single-name (risk weights 15–70% by bucket), emerging-market, and small-cap equity and commodities carry the largest marginal charges. The model predicts they migrate to unconstrained investors and earn the funding-plus-regulatory premium of Proposition 3. The clientele prediction is testable in holdings data. Bank-to-nonbank holding ratios should decline in marginal capital intensity, conditional on fundamentals.

5.3 Calibrating the pricing hit on regulated exposures

How large are these effects? Equation (26) gives the quantity to calibrate. Relative to an otherwise-identical zero-charge exposure, asset i must earn an additional expected return of

$$\text{pricing hit}_i = \chi\lambda \times \underbrace{\frac{\partial\mathcal{K}/\partial\theta_i}{p_i}}_{\text{marginal capital intensity}}, \quad (36)$$

the product of an economy-wide tightness statistic and an asset-specific intensity that can be read off the FRTB risk weights. The two pieces can be disciplined separately.

The tightness statistic $\chi\lambda$. A plausible range for the share χ of risk-bearing capacity held

by constrained intermediaries is one quarter to one half in dealer-intermediated fixed-income, FX, and derivatives markets, where standardized-approach banks are the marginal investor, and less in equities. For λ , the annual excess cost of a dollar of capital tied to the requirement, three routes give a range. These are banks’ required equity spread over the riskless rate (roughly 8–12%), the revealed-preference estimates of [Kisin and Manela \(2016\)](#), which imply materially lower values in calm periods, and stress-period wedges such as covered interest parity deviations ([Du et al. 2018](#)), which imply higher ones. Taking $\lambda \in [3\%, 12\%]$ and $\chi \in [1/4, 1/2]$ brackets $\chi\lambda$ between roughly 1% and 5%, with 3% a reasonable central value. Since the pricing hits below depend on χ and λ only through their product $\chi\lambda$, the calibration is robust to the split between them. Propositions 5 and 6 imply that the statistic is countercyclical, so the hits below should be read as widening in stress.

Marginal capital intensity. For a directional position, intensity is approximately the risk weight times the relevant duration (GIRR, CSR) or the risk weight itself (equity, FX, commodity), divided by $\sqrt{2}$ where the standard allows it for liquid currencies and pairs.⁶ Table S.2 in the supplementary appendix reports intensities for eleven representative exposures using the delta risk weights of the standard, together with the implied pricing hits across the $\chi\lambda$ range. All figures are medium-scenario, delta-only, and ignore the MAR21.6 scenario-max, so they are conservative.

Three observations follow. First, the magnitudes are in the range of documented post-crisis pricing wedges. At the central $\chi\lambda = 3\%$, the model implies roughly 20 basis points on ten-year duration and 40 on thirty-year, 50 on five-year investment-grade corporate credit, 30–45 on FX and equity-index exposure, and 90–165 on single-name and emerging-market equity. These are the same order as covered interest parity deviations (20–40bp), negative long-end swap spreads, and funding value adjustments ([Du et al. 2018](#); [Jermann 2020](#); [Andersen et al. 2019](#)). The model reads these not as anomalies of particular markets but as one cross-section indexed by marginal capital intensity. Second, the cross-sectional spread is the identifying prediction. The same $\chi\lambda$ that prices ten-year duration at 20bp prices the cross-curve *twin* of that duration at less than 1bp. The result is a 22-fold gap in required returns between positions with comparable true risk, which is the gap of Corollary 3 evaluated at the standard’s risk weights. Third, because every entry scales with $\chi\lambda$, the entire table widens and narrows with dealer capital tightness, giving the framework a single time-series factor and a full cross-section of exposures on which to test it.

5.4 An observable measure of the distortion

The tilt portfolio of Proposition 7 is the model’s sufficient statistic for the requirement’s effect on prices, but it is not directly observable, since it is built from equilibrium prices and the true covariance matrix. Its image in the space of *reported sensitivities*, however, is observable, and this

⁶The divisor is the square-root-of-time rule applied to the standard’s own liquidity assumption. Delta risk weights are calibrated as stressed factor moves over a liquidity horizon, and for interest rate risk in a specified list of currencies (MAR21.44) and for the most liquid currency pairs (MAR21.87) the standard permits the weight to be divided by $\sqrt{2}$, reflecting a horizon of ten rather than twenty days. The concession is a liquidity judgment rather than a volatility measurement: it lowers the calibration wedge for liquid-currency duration and FX relative to other exposures with no change in true risk.

subsection defines that diagnostic; the supplementary appendix shows how a supervisor would build it. Recall that $\chi = m_C \bar{\eta} / \eta_C$ is the constrained sector's share of aggregate risk-bearing capacity, so that in the frictionless benchmark the banking sector in aggregate holds $\chi \bar{\theta}$, the fraction χ of the market portfolio.

Definition 2 (Sensitivity gap). Let $s^{\text{bank}} := M'_r(m_C \theta^C)$ denote the aggregate net regulated sensitivities of the banking sector and $s^{\text{mkt}} := M'_r \bar{\theta}$ those of the market portfolio. The *sensitivity gap* is the vector of weighted-sensitivity shortfalls

$$g := D_w \left(\chi s^{\text{mkt}} - s^{\text{bank}} \right), \quad G := \|g\|, \quad (37)$$

the factor-by-factor difference, in the standard's own capital units, between the regulated sensitivities the banking sector would report in the frictionless benchmark and those it actually reports.

Proposition 8 (The gap measures the distortion). *In any binding equilibrium,*

$$g = \chi(1 - \chi) \frac{\lambda}{\bar{\eta}} D_w M'_r \tau, \quad \text{while} \quad \pi - \bar{\eta} \Sigma \bar{\theta} = \chi \lambda \Sigma \tau, \quad (38)$$

where τ is the tilt portfolio of Proposition 7. The gap and the vector of pricing distortions are linear images of the same equilibrium object $\lambda \tau$, so the gap vanishes when the requirement is slack and is generically nonzero whenever it binds.

The proof follows from Proposition 7, which gives $m_C \theta^C - \chi \bar{\theta} = -\chi(1 - \chi)(\lambda / \bar{\eta}) \tau$, and from the pricing equation (26). The content is in what the two expressions in (38) do *not* share. The pricing distortion is filtered through the unobservable Σ , while the gap is filtered through the regulator's own $D_w M'_r$, which is the map banks already apply to their books when they compute the charge. Unlike the shadow cost, the gap cannot fall while the distortion rises, because both are increasing images of the same tilt. When banks respond to high weights by shedding the charged book, λ collapses but the shed quantity, which is what g records, keeps growing. Section 6.6 verifies this in the calibrated economy, where under a proportional increase in risk weights G rises monotonically with the largest pricing hit through the entire region in which λ peaks and falls.

Both ingredients of (37) are computable from data the regime already generates. The bank term is the sum across standardized-approach banks of the weighted sensitivities $WS_k = w_k s_k$ that are line items of the SBM aggregation itself (MAR21.4), and the benchmark term needs only the regulated-factor sensitivities of the outstanding market portfolio together with an estimate of the intermediary share χ , both of which move slowly. The supplementary appendix shows how a supervisor assembles the indicator factor by factor and reports an illustrative dashboard for the calibrated economy (Table S.1). A companion market-based indicator is the family of coverage-basis spreads of Section 5.2, which reflect the same tilt in prices rather than in quantities.

5.5 The welfare cost of the requirement and its incidence

The closed-form value functions permit a welfare accounting. Welfare for class j is the certainty equivalent of date-1 wealth, $R_f W_0^j + V^j(\theta^j)$, compared between the binding equilibrium and the frictionless benchmark. The supply side enters as well, through the date-0 proceeds $p'\bar{\theta}$ received by the sellers of the fixed supply.

Proposition 9 (Welfare accounting). *Let $(p, \pi, \theta^U, \theta^C)$ be a binding equilibrium with multiplier $\lambda > 0$, let $\pi^0 = \bar{\eta} \Sigma \bar{\theta}$ denote the frictionless premia, and let τ be the tilt portfolio (33). Write $\|x\|_\Sigma^2 := x' \Sigma x$ and $T := \bar{\theta}' [\psi D_\zeta p + \nabla \mathcal{K}(\theta^C)]$. Relative to the frictionless equilibrium:*

- (i) *Traditional investors gain $\Delta CE^U = \frac{\chi\lambda}{\eta_U} [\bar{\eta} T + \frac{\chi\lambda}{2} \|\tau\|_\Sigma^2]$, which is positive whenever $T \geq 0$.*
- (ii) *The constrained investors' change is a scarcity gain net of a compliance cost,*

$$\Delta CE^C = \frac{\chi\lambda}{\eta_C} \left[\bar{\eta} T + \frac{\chi\lambda}{2} \|\tau\|_\Sigma^2 \right] - \frac{\lambda^2}{2\eta_C} \|\tau\|_\Sigma^2,$$

and its sign is ambiguous, positive when λ is small relative to T and negative when the requirement is tight.

- (iii) *The supply side's date-0 proceeds fall by $\chi\lambda T/R_f$.*
- (iv) *Summing (i)–(iii) at population weights and the riskless rate, every price term cancels as a transfer, and total surplus falls by the deadweight loss*

$$\text{DWL} = \frac{\chi(1-\chi)}{2\bar{\eta}} \lambda^2 \|\tau\|_\Sigma^2 \geq 0. \quad (39)$$

Two points stand out. First, the deadweight loss is *second order* in the shadow cost, growing as λ^2 while every pricing effect in Section 4 is first order in $\chi\lambda$, so price distortions are not themselves welfare statements. Second, the incidence is uneven. The unconstrained sector always gains, the constrained sector can gain as well when the scarcity premium outweighs its compliance cost, and the burden falls on the supply side through lower date-0 prices, that is, on issuers' cost of capital.

The loss also sits on a different margin from the one that cost-benefit assessments of capital usually weigh. Those exercises (Basel Committee on Banking Supervision 2010; Firestone et al. 2019) price the *level* of capital, its funding and output costs against the crises it prevents. Equation (39) is instead a *composition* cost, present at any level of capital and set by the structure of the charge rather than its quantity, so a regulator who has already fixed the level optimally can still reduce it by recalibrating relative weights, at no cost in loss absorbency. The accounting here prices this cost margin alone, and says nothing about the default and stability benefits the model deliberately omits, against which (39) would be weighed.

6 Numerical Illustration

This section solves the model at FRTB parameters, using the scalar reduction of Proposition 4. For each candidate (λ, κ) and sign configuration of banks' books the economy is linear and solves in closed form, and the equilibrium is the root of the two consistency conditions (30), a two-dimensional problem regardless of the number of assets and factors. Under a self-financed book ($\psi = 1$) the gross funding term has a kink wherever banks' position in an asset is zero, and the solver enumerates all $3^N = 81$ sign configurations, keeping the one whose solution is consistent with it. Under a pure capital requirement ($\psi = 0$) there is no funding term and a single two-dimensional problem remains. At every reported solution, all of the paper's equilibrium identities (market clearing, both first-order conditions, the pricing equation of Proposition 2, the tilt identity of Proposition 7, and the welfare decomposition of Proposition 9) are verified numerically to within 10^{-11} .⁷ Every result is reported at both endpoints of the financing parameter.

6.1 Calibration

The economy has $N = 4$ assets and $L = 4$ factors, of which $K = 3$ are regulated. Table S.3 in the supplementary appendix summarizes the calibrated parameters. The first asset is ten-year government duration in a liquid currency (GIRR, vertex risk weight $1.1\%/\sqrt{2}$, duration 8.5, capital intensity 6.61% per dollar). The second is an advanced-economy equity index (equity bucket 12, risk weight 15%), and the third is a large-cap advanced-economy equity position (bucket 5, risk weight 30%). The fourth asset loads on the one unregulated factor, f_4 , which lies outside the regulator's list. It has 5.5% systematic volatility and is positively correlated with the equity factors, but its capital charge is zero because the SBM does not measure its factor. It is the model's stand-in for the exposures Section 5.2 identifies as outside the factor list. The true index/large-cap factor correlation is 0.85, against a prescribed cross-bucket $\gamma = 0.45$ from MAR21.80, so both wedges are present in the calibration. The cross-risk-class correlation in Γ is set to zero.⁸ Investors have equal risk aversion ($\eta_U = \eta_C = 0.05$) and banks have total mass $m_C = 1/3$, so $\chi = 1/3$, and $R_f = 1.04$. Expected payoffs are set so that frictionless prices equal one, delivering frictionless premia of 1.05% (duration), 5.0–5.6% (equity), and 0.94% (uncharged asset).

The binding threshold of Corollary 1 is $\bar{W} = 111.2$ under a self-financed book and $\bar{W} = 7.2$, the charge on the frictionless bank book, under a pure capital requirement, so comparable tightness across the endpoints cannot be had by holding bank equity fixed. The baseline endows banks with $W_0^C = 60$ at $\psi = 1$, about 55% of the threshold, and with $W_0^C = 6.6$ at $\psi = 0$, about 92% of

⁷The enumeration doubles as a multiplicity check. For each configuration the budget condition is scanned on a 240-point grid in λ , every sign change and every near-tangency is bracketed, and forty Newton starts are added; at the baseline and in every stress scenario exactly one configuration admits a consistent solution, and it has a single root. Under a pure capital requirement, where the configuration is immaterial, a probe from 200 dispersed starting points in (λ, κ) finds the reported equilibrium and no other.

⁸The actual standard *sums* the risk-class charges, which corresponds to a higher effective cross-class correlation for same-sign books. Setting it to zero gives banks cross-class diversification that the rule does not recognize, so the regulatory differentials below are conservative.

Table 1: Baseline equilibrium at both endpoints of the financing parameter. Panel A: self-financed book, $W_0^C = 60$, $\lambda = 0.997\%$, $\chi\lambda = 33.2\text{bp}$, $\kappa = 5.44$. Panel B: pure capital requirement, $W_0^C = 6.6$, $\lambda = 3.14\%$, $\chi\lambda = 104.6\text{bp}$, $\kappa = 6.60$. “Total” is the equilibrium expected excess return less its frictionless value, and it splits into three additive parts: the funding wedge $\psi\chi\lambda\varsigma_i$, the regulatory term of (36), and a fundamental part that is true risk repriced at the higher equilibrium discount rate. The bank share is $m_C\theta_i^C/\bar{\theta}_i$, with frictionless value 1/3. ς_i is the funding subgradient of (19), equal to one when banks are long and interior when they sit at the kink. Displayed components need not sum to the total because of rounding.

Asset	$\mathbb{E}[R_i] - R_f$ (%)		Pricing hit (bp)				Bank	
	frictionless	equilibrium	total	funding	regulatory	fundamental	share	ς_i
<i>Panel A: self-financed book ($\psi = 1$)</i>								
10y govt bond	1.05	1.39	34.1	33.2	0.6	0.3	0.12	1.00
Equity index	5.02	5.42	39.5	33.2	4.5	1.8	0.29	1.00
Large-cap equity	5.61	6.03	42.7	33.2	7.3	2.2	0.31	1.00
Uncharged asset	0.94	1.04	10.1	10.0	0.0	0.1	0.00	0.30
<i>Panel B: pure capital requirement ($\psi = 0$)</i>								
10y govt bond	1.05	1.09	3.9	0.0	3.9	0.0	0.31	—
Equity index	5.02	5.15	13.0	0.0	12.4	0.6	0.32	—
Large-cap equity	5.61	5.80	19.3	0.0	18.3	1.0	0.26	—
Uncharged asset	0.94	0.94	0.0	0.0	0.0	0.0	0.54	—

it, the latter chosen so that the tightness statistic $\chi\lambda$ lands at one percent, the bottom of the back-of-the-envelope range of Section 5.3.

6.2 The baseline equilibrium

Table 1 reports the solution at both endpoints. Under a self-financed book the shadow cost is $\lambda = 1.00\%$ per year, so the tightness statistic is $\chi\lambda = 33\text{bp}$. The funding wedge is 33bp on each of the three charged assets, which banks hold long, and 10bp on the uncharged asset, in which banks hold a zero position, sitting at the kink of (16). At this kink, the subgradient $\varsigma_4 = 0.30$ determined by market clearing means the asset bears three tenths of the wedge. The regulatory components on top are 0.6bp (duration), 4.5bp (index), 7.3bp (large-cap), and zero (uncharged asset), ordered by equilibrium marginal capital intensity as (36) requires. Under a pure capital requirement (Panel B) there is no funding wedge at all: $\chi\lambda = 105\text{bp}$, the entire pricing hit is regulatory, from 3.9 to 18.3bp on the charged assets and exactly zero on the uncharged one, and the security market line passes through R_f as Proposition 3 says it must.

Four features of the baseline deserve comment.

Where the uncharged asset ends up depends on the financing margin. Under a pure capital requirement banks hold 54% of the uncharged asset’s supply against a frictionless share of one third. Charge-free risk is the cheapest risk on their balance sheets, and it accumulates there as inventory, which is the pattern anticipated after Proposition 7. Under a self-financed book the same asset consumes own resources like any other, its premium at the baseline calibration does not

cover that cost, and banks hold none of it. Neither endpoint lets banks use the uncharged asset as a source of regulatory capacity, since under gross funding a short consumes what a long consumes. Section 6.4 shows that this margin, not the factor list, governs the size of the measured coverage basis.

The calibration wedge is large where the coverage wedge is not. A hedged-equity portfolio (long \$1 large-cap, short \$1 index) has true volatility of 11.7% per dollar long, but a capital charge of 26.8 cents, which is 2.3 times its true risk. The reason is that offset depends on correlation. At the true correlation of 0.85 the two legs cancel most of each other’s risk, but the charge is computed at the prescribed cross-bucket correlation of 0.45 (MAR21.80), at which they cancel much less. At the mild baseline tightness this costs the hedge only 2.8bp of equilibrium premium at $\psi = 1$ and 5.9bp at $\psi = 0$, but the charge ratio is invariant to λ , so in stress the same book pays 2.3 times its risk-proportionate share of any tightening.

At statutory-scale weights, coverage shapes composition more than measured tightness. Recomputing the equilibrium with the coverage wedge closed, meaning the unregulated factor is charged neutrally at its true volatility so that $K = L$, leaves λ unchanged to three decimal places at $\psi = 1$ and moves it from 3.14% to 3.36% at $\psi = 0$. The coverage wedge’s first-order effect at baseline is on *who holds what* and on relative prices, and only secondarily on the shadow cost a supervisor would measure.

The welfare cost is small relative to the effect on prices, and banks bear none of it. Evaluating Proposition 9 at the baseline gives a deadweight loss of 7.4 basis points of aggregate market value per year under a self-financed book, about a fifth of the 34–43bp pricing hits, and 0.3bp under a pure capital requirement, an order of magnitude below its hits, as the second-order formula (39) requires. At $\psi = 1$ traditional investors gain 25.1bp, the supply side loses 34.0bp through lower date-0 prices, and banks gain 1.5bp, because at baseline tightness the scarcity gain exceeds the compliance cost, so the regulated sector is a net beneficiary of its own requirement; the same signs obtain at $\psi = 0$. Closing the coverage wedge leaves the loss essentially unchanged at either endpoint, the welfare counterpart of the composition finding above.

6.3 Stress and the impairment of the unconstrained sector

The distortions above are computed in an economy whose unconstrained investors stand ready to absorb whatever banks shed. Stress episodes are the opposite configuration. Dealer capital falls at the same time as the risk-bearing capacity of nonbank investors, which is the pattern of March 2020 in dealer-intermediated markets. Becker and Benmelech (2025) document the behavior of the United States corporate bond market in that episode. In March 2020 the two impairments arrived together: dealers’ balance sheets were already full when the mutual funds and foreign holders who would normally absorb what dealers shed became sellers themselves, so the marginal buyer of what dealers could not warehouse was scarce on both sides of the market. The joint-stress scenario below is the model’s version of that scenario. Table 2 recomputes the equilibrium under two impairments,

Table 2: Stress scenarios at both endpoints of the financing parameter. Each scenario is solved at baseline fundamentals with the indicated preferences and bank equity. Hits are measured against the scenario’s own frictionless benchmark. The coverage spread is the pricing-hit gap between the most capital-intensive asset (large-cap equity) and the uncharged asset. The last column is banks’ position in the uncharged asset as a share of its supply.

Scenario	η_U	W_0^C	$\bar{W}$	λ	$\chi\lambda$ (bp)	Largest hit (bp)	Coverage spread (bp)	Uncharged share
<i>Panel A: self-financed book ($\psi = 1$)</i>								
Baseline	0.05	60	111.2	1.00%	33	43	32.5	0.00
Bank stress	0.05	30	111.2	2.22%	74	96	77.9	0.00
Nonbank impairment	0.15	60	196.5	4.85%	291	400	324.5	0.00
Joint stress	0.15	30	196.5	13.11%	786	1105	940.8	0.00
<i>Panel B: pure capital requirement ($\psi = 0$)</i>								
Baseline	0.05	6.6	7.2	3.14%	105	19	19.3	0.54
Bank stress	0.05	3.3	7.2	23.3%	778	123	123.5	1.91
Nonbank impairment	0.15	6.6	12.9	63.1%	3787	695	694.8	3.16
Joint stress	0.15	3.3	12.9	97.8%	5870	1071	1070.6	4.58

separately and together. These are bank equity halved and unconstrained risk-bearing capacity cut to a third (η_U raised from 0.05 to 0.15). Fundamentals (μ , Σ , supplies, risk weights) are held at their baseline values, and each scenario’s pricing hits are measured against that scenario’s own frictionless benchmark, so the reported distortions isolate the contribution of the capital requirement under the stressed conditions.⁹

Under a self-financed book, halving bank equity alone moves the tightness statistic from 33 to 74 basis points, which is painful but modest, because deep-pocketed unconstrained investors keep absorbing the tilt at little price concession. Impairing those investors raises both factors of $\chi\lambda$ at once. The banking sector’s share of the economy’s remaining risk-bearing capacity χ rises from one third to 0.60, and the shadow cost λ jumps to 4.9% because the residual investors now demand far more compensation to hold what banks shed. The tightness statistic reaches 291 basis points, and 786 in the joint scenario, with the largest pricing hit at 400 basis points under nonbank impairment and 1105 under joint stress. A pure capital requirement transmits the same impairments far more intensely, because banks’ only cushion is the charge itself and the stressed equity levels sit at a quarter to a half of a threshold that was already nearly binding. The absolute levels in Panel B under stress are outside the range in which a static, delta-only calibration should be read literally, and it is the ordering and the cross-section that carry over.

Two interesting things come from the stressed case considered here. First, the coverage spread widens with tightness at both endpoints, from 33 basis points at baseline to 941 under joint stress at $\psi = 1$, a factor of 29, and from 19 to 1071 at $\psi = 0$. The model’s most identifiable prediction, that on-list/off-list spreads between comparably risky exposures comove with intermediation capacity,

⁹Each stressed equilibrium is computed from scratch by the configuration enumeration described at the start of this section, so no continuation or warm start is involved, and in each case exactly one configuration admits a consistent solution. All equilibrium identities are verified at the reported solutions.

is most visible in stress, just when the corresponding empirical objects (sudden widening of such spreads, dealer balance-sheet congestion) are observed. Second, the location of uncharged risk again tracks the financing margin. Under a self-financed book banks stay at the kink in every scenario and hold none of the uncharged asset, while under a pure capital requirement their inventory grows from 54% of supply to 458% under joint stress, because charge-free risk is most valuable to a constrained balance sheet when capital is scarcest. The sensitivity gap of Section 5.4 registers the shedding of charged risk in both cases, while the headroom-based indicators a supervisor conventionally watches would register only the movement in λ .¹⁰

6.4 Financing, the off-list Sharpe ratio, and the coverage basis

Corollary 3 says an off-list exposure trades at a discount to an on-list twin. Whether that discount appears in a calibrated economy turns out to depend on something the corollary holds fixed: the reward the off-list exposure offers on its own, and hence whether banks want to hold it at all. This subsection varies the systematic volatility σ_4 of the unregulated factor, which moves the off-list asset's frictionless Sharpe ratio, and reports the equilibrium at both ends of the financing parameter. Table 3 collects the results. Tightness is held comparable across columns by fixing the ratio of bank equity to its binding threshold at its baseline value rather than fixing the level of equity.

Under a self-financed book, banks hold the off-list asset only if its premium covers its funding cost. In Panel A banks hold *none* of the off-list asset until its Sharpe ratio reaches roughly 0.20. The reason is visible in the funding-sign column. Holding the asset long consumes ψp_4 of own resources per share and earns no offsetting reduction in the charge, so banks take a position only once the asset's premium covers that funding cost. Below the threshold banks sit exactly at the kink of (16), and the subgradient ς_4 is interior: 0.30 at the paper's baseline calibration, rising to 0.69 and then to 1.00 as the asset becomes more rewarding. The off-list asset bears the fraction ς_4 of the funding wedge rather than all of it or none of it, and market clearing, not the capital rule, is what sets that fraction.

This makes the measured basis at $\psi = 1$ a partly spurious object. Reading down the last column of Panel A, the basis falls from 32.5 to 8.3 basis points as σ_4 rises, a factor of four, while the risk weights, the correlation matrix, the factor list, and the coverage wedge are all held fixed. Nothing about regulatory coverage has changed across those rows. What changes is how much of the *funding* wedge the off-list asset happens to bear, which is an artifact of where banks sit relative to the kink. At the top of the panel the off-list asset bears only 30% of a wedge that has nothing to do with the factor list, and the resulting gap is recorded as a coverage basis. At the bottom of the panel, where banks hold the asset at nearly its frictionless share, the basis collapses to the 8–9 basis points that

¹⁰The welfare accounting under stress: at $\psi = 1$ the deadweight loss (39) rises from 7 basis points of market value at baseline to 21 under bank stress and 176 under joint stress, about a third of the aggregate risk premium. Banks' own position changes sign only under bank stress, where the compliance cost overtakes the scarcity gain and banks lose 6.6bp on net; impairing nonbank investors raises the scarcity premium enough that banks gain even in the joint scenario. The same pattern obtains at $\psi = 0$.

Table 3: The off-list asset’s Sharpe ratio, banks’ position in it, and the coverage basis. Each row varies the systematic volatility σ_4 of the unregulated factor, holding the regulated factors, the risk weights, and $W_0^C/\bar{W}$ fixed at the baseline (0.54 at $\psi = 1$, 0.92 at $\psi = 0$). “Bank share” is $m_C\theta_4^C/\bar{\theta}_4$, with frictionless value $\chi = 1/3$. “Funding sign” is the subgradient ς_4 of (19), equal to one when banks are strictly long and interior when they sit at the kink. “Basis” is the pricing-hit gap between the large-cap equity position and the off-list asset.

σ_4	$\chi\lambda$ (bp)	Off-list asset		Bank		Pricing hit (bp)	
		premium	Sharpe	share	funding sign	off-list	basis
<i>Panel A: $\psi = 1$, self-financed book</i>							
0.055	33.2	0.94%	0.168	0.00	0.30	10.1	32.5
0.100	33.9	1.85%	0.184	0.00	0.69	23.9	19.7
0.150	35.5	2.99%	0.199	0.12	1.00	36.5	9.2
0.200	36.5	4.29%	0.214	0.26	1.00	38.0	9.1
0.300	36.8	7.33%	0.244	0.34	1.00	39.4	8.3
<i>Panel B: $\psi = 0$, pure capital requirement</i>							
0.055	104.6	0.94%	0.168	0.54	—	0.0	19.3
0.100	104.2	1.85%	0.184	0.45	—	0.0	19.2
0.150	104.1	2.99%	0.199	0.41	—	0.0	19.2
0.200	104.0	4.29%	0.214	0.39	—	0.0	19.2
0.300	104.0	7.33%	0.244	0.37	—	0.0	19.3

the difference in *charges* actually warrants. A supervisor or econometrician measuring the basis under this specification would be measuring the financing assumption as much as the rule.

Under a pure capital requirement the basis is a clean measure of coverage. Panel B is invariant. The off-list asset bears a pricing hit of exactly zero at every Sharpe ratio, which is what Corollary 2 predicts: a portfolio in the null space of M_r' consumes no own resources at $\psi = 0$ and is therefore priced as though the requirement did not exist. The basis is then simply the regulatory premium on the charged twin, 19.2 basis points, and it does not move when the off-list asset’s own risk and reward change. Banks hold the asset long in every row, and hold *more* of it when its Sharpe ratio is low, because banks’ absolute off-list position is bounded by risk aversion rather than by the constraint, so a smaller off-list factor translates into a larger share of its supply. As σ_4 rises the share converges towards the frictionless $\chi = 1/3$ from above.

Two implications. The first is a modeling one. If the coverage basis is to serve as the paper’s identifying prediction, it should be derived in the specification where it measures coverage and nothing else. Panel A’s basis inherits the financing assumption, and its magnitude at the baseline calibration is an accident of banks sitting at a kink. Panel B’s does not. The second is empirical, and sharper. The model with $\psi = 0$ predicts not merely that on-list and off-list twins price differently, but that the size of the gap is *independent of the off-list exposure’s own risk-return profile*, since it is set entirely by the charge on the on-list leg. That is an auxiliary restriction a test can reject. A basis that widens or narrows systematically with the off-list leg’s volatility or Sharpe ratio would point away from a coverage explanation and towards a funding or inventory-risk one.

A final caveat on Panel A. The sensitivity to the financing assumption runs in both directions.

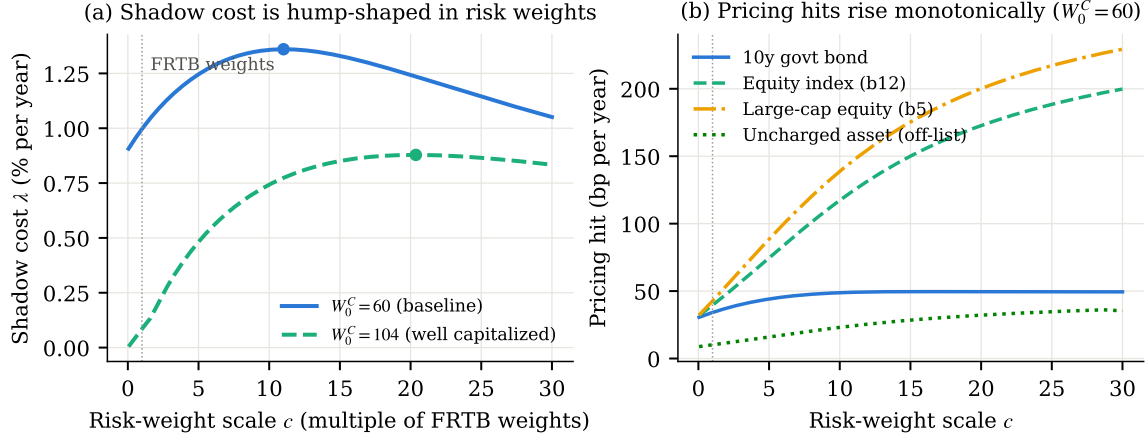


Figure 1: A proportional change in all risk weights, each scaled by the common factor c , under a self-financed book ($\psi = 1$). Panel (a): the shadow cost is hump-shaped (peaks marked), at both baseline and high bank equity. Panel (b): pricing hits at baseline equity rise monotonically throughout. Past the peak of λ , prices keep deteriorating while the measured shadow cost falls. Figure S.2 in the supplementary appendix repeats the exercise under a pure capital requirement.

Under the netting alternative $\psi p'\theta$ discussed in Section 3.3, which lets shorts create capital space, banks in the baseline calibration take a short position of -121% of the off-list asset's supply and the basis reads 5.4 basis points; under gross funding at the same parameters they hold zero and the basis reads 32.5; under a pure capital requirement they hold $+54\%$ and the basis reads 19.3. Three specifications of the financing margin, one unchanged capital rule, and a sixfold range in the object the paper proposes to measure. The financing margin is not a modeling detail here, and it is the reason the results are reported at both ends of ψ throughout.

6.5 Comparative statics at FRTB parameters

Figure 1 scales every risk weight by a common factor c and is the central exercise of this section.¹¹ The shadow cost is *hump-shaped*, peaking at $c \approx 11$ at baseline equity and declining thereafter, and peaking near $c \approx 20$ at high bank equity ($W_0^C = 104$, just enough to fund the frictionless book with nothing left for the charge). Yet the pricing hits rise *monotonically* throughout, reaching roughly 200–230bp on the equity assets at $c = 30$ while duration and the uncharged asset stabilize near 35–50bp (their residual hit is mostly the funding wedge). This confirms, in the multi-asset general equilibrium, the hump in the shadow cost and the monotone rise in the premium distortion that Proposition 6(iii) derives for the single-factor economy. The quantities move in the same way, with banks' share of the charged assets collapsing (from 12–31% of supply at $c = 1$ to 2–10% at $c = 30$). The uncharged asset is where the kink matters: banks hold none of it across almost the entire

¹¹The companion exercise, lowering bank capital at fixed weights, is reported in Figure S.1 of the supplementary appendix. Under a self-financed book the shadow cost rises from zero at $\bar{W}$ to 4.5% at $W_0^C = 15$, the equity hits reach 180–200bp, and the spread between the most and least capital-intensive assets widens from 33bp to 164bp, the model's prediction that these price gaps comove with dealer capital in general-equilibrium units. Along the way banks exit duration entirely, at about $W_0^C = 30$.

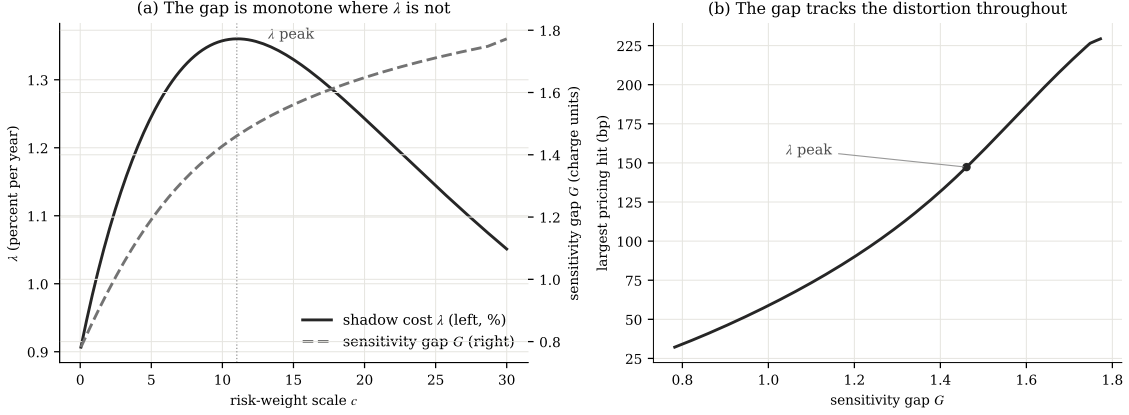


Figure 2: The sensitivity gap under the proportional change in risk weights of Figure 1, self-financed book. Panel (a): the shadow cost λ (solid, left axis) is hump-shaped with its peak at $c \approx 11$. The sensitivity gap G of Definition 2 (dashed, right axis) rises monotonically throughout. Panel (b): the largest pricing hit plotted against G . The dot marks the λ peak, which is invisible in the relationship. To keep units comparable across c , G is evaluated at the statutory ($c = 1$) weights. Evaluating it at the scaled weights cD_w only steepens the profile. All identities verified at every grid point. Figure S.3 in the supplementary appendix repeats the exercise under a pure capital requirement.

range, but the subgradient ς_4 climbs from 0.30 at $c = 1$ to 0.90 at $c = 25$ as high weights price banks out of regulated risk, and at $c = 30$ the position leaves the kink and banks go long. Under a pure capital requirement (Figure S.2) the same hump appears earlier and higher, with λ peaking at 11.7% near $c \approx 2$ while the equity hits climb monotonically to 190–220bp, and banks’ inventory of the uncharged asset rises from 54% of supply to 325%.

Consider two useful takeaways from this calibration. First, under a self-financed book the equilibrium tightness statistic is modest at plausible parameters ($\chi\lambda$ stays in the 33–45bp range as risk weights are scaled up) because prices adjust, banks shrink their books, and the taxonomy’s uncharged margins absorb part of the burden; reaching the large end of the range of Section 5.3 requires either a pure capital requirement or impairing *unconstrained* investors’ capacity, the experiment of Section 6.3. Second, the cross-section, not the level, is where the framework’s distinctive predictions are clearest. The ordering of the regulatory components, the hedge-penalty ratio, and the migration of charged risk out of the constrained sector are sign-definite implications that survive at every point of both exercises, at both endpoints of ψ .

6.6 The sensitivity gap as risk weights rise, and a recalibration counterfactual

Two final exercises connect the model’s supervisory content to the calibrated economy. The first is the sensitivity gap of Section 5.4 computed as risk weights are scaled up, and the second is the recalibration counterfactual promised by the wedge decomposition (10).

The sensitivity gap tracks the distortion where the shadow cost does not. Figure 2 recomputes

the risk-weight scaling of Figure 1 and reports the gap G alongside λ and the largest pricing hit. Proposition 8 says that the gap and the pricing distortion are both proportional to the product of the shadow cost and the tilt, so they must move together. The figure shows that they do. The shadow cost peaks at $c \approx 11$ and declines thereafter, while G rises monotonically from 0.87 at statutory weights to 1.77 at $c = 30$, rising with the pricing hits throughout. Panel (b) shows the hit as a strictly increasing function of G along the entire range of risk weights, with the λ peak an unremarkable interior point. Under a pure capital requirement (Figure S.3) the contrast is starker still: λ peaks at $c \approx 2$ and has fallen by almost nine tenths by $c = 30$, while G climbs from 0.21 to 2.09 and the largest hit keeps rising. A supervisor plotting reported weighted sensitivities against a market benchmark would see the requirement’s effect on prices growing throughout the region in which every headroom-based indicator reports it fading.

Recalibration repairs the calibration wedge, at essentially zero cost in measured tightness, and cannot touch the coverage wedge. The counterfactual replaces the statutory parameters with the neutral calibration of Section 3.2, with risk weights proportional to the true volatilities of the three regulated factors, prescribed correlations equal to the true ones, and the scale ϕ chosen so that the charge on the frictionless bank book is unchanged ($\phi = 1.01$, delivering weights of 6.0%, 16.2%, and 19.2% against the statutory 6.6%, 15%, and 30%). At baseline equity under a self-financed book, measured tightness barely moves ($\chi\lambda$ goes from 33.2 to 33.3 basis points), while the charge on the dollar-matched hedged-equity book of Section 6.2 falls from 2.3 times its true risk to 0.86 (slightly below parity because the neutral weights charge systematic risk only). Under the joint-stress scenario of Table 2, where composition matters most, the regulatory spread among the charged assets narrows by a quarter, from 217 to 163 basis points, and the largest pricing hit falls from 1105 to 1031. Under a pure capital requirement the baseline effects are of the same kind, while under joint stress, where banks have already shed nearly all charged risk, reweighting leaves the stressed cross-section essentially where it was. What does not move, in any scenario, is the coverage margin: banks’ position in the uncharged asset and the baseline deadweight loss are essentially unchanged with either set of weights. Reweighting removes the distortions the weights create and leaves untouched the ones the factor list creates.

7 Conclusion

The sensitivities-based method makes the covariance matrix a regulatory instrument. This paper works out what that choice does to asset prices in a transparent equilibrium model. Expected returns follow a two-beta CAPM whose second beta is computed under the regulator’s matrix, the wedge between that matrix and the true one splits into a calibration component the regulator can remove and a coverage component it cannot, constrained intermediaries tilt toward uncovered risk, and a basis opens between otherwise identical twin exposures. The shadow cost of the requirement is hump-shaped in the rule’s stringency while the pricing distortions rise monotonically, so the indicators that track the shadow cost understate the requirement where its effect on prices is

largest, and the sensitivity gap corrects this from data the regime already collects.

This has implications for the officials who calibrate and supervise the rule. First, quantities should be monitored along with shadow prices. The sensitivity gap of Definition 2 is, by Proposition 8, a linear image of the same equilibrium object that drives every pricing distortion, its inputs are the weighted sensitivities that enter the SBM calculation itself, and Figure 2 shows it rising monotonically through the entire region in which the shadow cost peaks and falls. The family of coverage-basis spreads (on-/off-the-run, cash-futures, bond-CDS) is a market-based complement that is observable daily. Second, a change in a delta risk weight moves relative prices economy-wide in proportion to $\chi\lambda$, with incidence on issuers and unconstrained investors as well as on banks, so calibration reviews can be scored by their effect on the cross-section of expected returns and not only on capital ratios. The decomposition (10) gives such reviews a concrete objective, since the calibration component is removable without changing the aggregate stringency of the regime; at the standard's cross-bucket correlation of 0.45, against a true correlation of 0.85, a dollar-matched long-short equity book is charged 2.3 times its true risk. Third, the coverage margin is the harder problem. Whatever the weights, risk outside the prescribed factor list is free on constrained balance sheets and will migrate there, so the taxonomy is a decision about which risks the regulated sector will be induced to accumulate. The residual risk add-on, the framework's existing patch, levies flat percentages of notional and so reproduces the same invariance within its coverage; making it sensitive to even a crude risk proxy would blunt it, and sustained concentration of standardized-approach banks in instruments with low marginal charges relative to their volatility is detectable in the same supervisory sensitivity data. None of this weighs against capital adequacy as such. The model holds the level of capital fixed and shows that the structure of the charge, its relative weights, its prescribed correlations, and its coverage, determines a pattern of relative prices and risk locations that is currently unmonitored.

Further research in this area could fill in several important questions. A dynamic version would let the procyclicality of $\chi\lambda$ interact with capital accumulation, and adding the vega, curvature, and default-risk layers, and the scenario-max layer of MAR21.6, would bring the model closer to the standard as written. One open empirical question is whether on-list and off-list spreads between comparably risky exposures move together with dealer capital. Section 6.4 offers a way to test the coverage explanation. If it is right, the basis depends only on the charge on the on-list leg, so it should not change when the off-list leg's own risk or reward changes. The broader point is that when the marginal investor's capital depends on a risk model the regulator writes, differences between that model and underlying risk will affect asset prices.

Supplementary Appendix

This supplementary appendix states and proves the existence theorem, collects the proofs of all results in the main text together with one result stated here in full (Corollary S.1), and adds a section on assembling the sensitivity gap from supervisory data, three supplementary figures, and supplementary tables. Unprefixed numbers for lemmas, propositions, theorems, corollaries, equations, sections, figures, and assumptions refer to the main body. Items numbered with the prefix S (Corollary S.1, Figure S.1, Table S.1, and so on) are internal to this appendix.

Existence of equilibrium

This section states and proves the existence result referred to in Section 4.4 of the main text.

Theorem S.1 (Existence). *Under Assumption 1, for any $W_0^C > 0$, $m_U \in (0, 1)$, and $\bar{\theta} \in \mathbb{R}_+^N$, an equilibrium exists, provided that when $\psi > 0$ prices are positive on the set of candidate bank books, in the sense that*

$$\mu_i > \frac{\eta_U}{m_U} [(\Sigma \bar{\theta})_i + m_C \sigma_i \rho^*] \quad \text{for every } i, \quad \sigma_i := \Sigma_{ii}^{1/2}, \quad (\text{S.1})$$

with ρ^* the bound (S.3) below. If (28) holds, it is the CAPM allocation with slack requirement. If (28) fails, an equilibrium exists and in every equilibrium the requirement binds, $\lambda > 0$.

The proof rests on one feature of the CARA–normal structure. Since traditional investors are unconstrained, their first-order condition holds in any equilibrium, and market clearing then expresses risk premia as an affine function of the bank sector’s book alone:

$$\pi(\theta) = \frac{\eta_U}{m_U} \Sigma (\bar{\theta} - m_C \theta), \quad (\text{S.2})$$

so an equilibrium is a fixed point of the bank’s best response to the prices implied by its own book. Because $\theta = 0$ is always feasible ($W_0^C > 0$) and delivers zero excess utility, any fixed point generates nonnegative surplus at the prices (S.2) it induces, yielding the a priori bound

$$\|\theta^C\|_\Sigma \leq \rho^* := \frac{(\eta_U/m_U) \|\bar{\theta}\|_\Sigma}{\eta_C/2 + \eta_U m_C/m_U}, \quad \|x\|_\Sigma := \sqrt{x' \Sigma x}, \quad (\text{S.3})$$

on every candidate equilibrium book, for every configuration of risk weights. Existence then follows from Brouwer’s theorem applied to the best response truncated to this ball. The bound says that the bank’s book is limited by fundamentals and preferences alone, however the regulator sets (w, Γ) . Condition (S.1) is needed only under gross funding, where it keeps the funding term convex in the bank’s book by keeping prices positive on the ball of candidate books; it is a restriction on expected payoffs relative to risk premia and holds comfortably in the calibration of Section 6.

Proof of Theorem S.1. Step 0: positive prices on the relevant ball. Let $\rho := \rho^* + \varepsilon$ with $\varepsilon > 0$ small enough that, when $\psi > 0$, (S.1) still holds with ρ in place of ρ^* (possible since the inequality is

strict), and let $B := \{\theta : \|\theta\|_\Sigma \leq \rho\}$. For $\theta \in B$ and every i , $|(\Sigma\theta)_i| = |e'_i \Sigma\theta| \leq \|e_i\|_\Sigma \|\theta\|_\Sigma \leq \sigma_i \rho$, so

$$R_f p_i(\theta) = \mu_i - \frac{\eta_U}{m_U} [(\Sigma\bar{\theta})_i - m_C(\Sigma\theta)_i] \geq \mu_i - \frac{\eta_U}{m_U} [(\Sigma\bar{\theta})_i + m_C\sigma_i\rho] > 0$$

for the price map $p(\cdot)$ defined in Step 1. Hence for $\theta \in B$ the function $g(\theta', \theta) := \psi \sum_i p_i(\theta) |\theta'_i| + \mathcal{K}(\theta') - W_0^C$ is convex and continuous in θ' , jointly continuous in (θ', θ) , and $g(0, \theta) = -W_0^C < 0$. When $\psi = 0$ all of this holds for every $\theta \in \mathbb{R}^N$ without (S.1).

Step 1: reformulation as a fixed point. For $\theta \in \mathbb{R}^N$ define

$$\pi(\theta) := \frac{\eta_U}{m_U} \Sigma (\bar{\theta} - m_C \theta), \quad p(\theta) := R_f^{-1} [\mu - \pi(\theta)],$$

and let $C(\theta) := \{\theta' \in \mathbb{R}^N : g(\theta', \theta) \leq 0\}$ and $V(\theta'; \theta) := \theta'^U \pi(\theta) - \frac{\eta_C}{2} \theta'' \Sigma \theta'$. For $\theta \in B$, $V(\cdot; \theta)$ is strictly concave and coercive and $C(\theta)$ is closed, convex, and nonempty ($0 \in C(\theta)$) by Step 0, so the best response

$$\text{BR}(\theta) := \arg \max_{\theta' \in C(\theta)} V(\theta'; \theta)$$

is well defined and single-valued on B . The claim is that (p, θ^U, θ^C) with $\theta^C \in B$ is an equilibrium if and only if $\theta^C = \text{BR}(\theta^C)$, with $p = p(\theta^C)$ and $\theta^U = (\bar{\theta} - m_C \theta^C)/m_U$. Indeed, in any equilibrium the traditional investors' first-order condition $\pi = \eta_U \Sigma \theta^U$ combined with clearing $\theta^U = (\bar{\theta} - m_C \theta^C)/m_U$ gives $\pi = \pi(\theta^C)$, so the bank optimizes against $p(\theta^C)$ and $\theta^C = \text{BR}(\theta^C)$. Conversely, a fixed point delivers bank optimality by construction, traditional-investor optimality since $\eta_U \Sigma \theta^U = \pi(\theta^C)$, and clearing by the definition of θ^U .

Step 2: a priori bound. Suppose $\theta^* = \text{BR}(\theta^*)$, or more generally that θ^* maximizes $V(\cdot; \theta^*)$ over any subset of $C(\theta^*)$ containing 0. Then $V(\theta^*; \theta^*) \geq V(0; \theta^*) = 0$, i.e.

$$\frac{\eta_C}{2} \|\theta^*\|_\Sigma^2 \leq \theta^{*U} \pi(\theta^*) = \frac{\eta_U}{m_U} [\theta^{*U} \Sigma \bar{\theta} - m_C \|\theta^*\|_\Sigma^2].$$

Collecting terms and applying the Cauchy–Schwarz inequality in the inner product $\langle x, y \rangle_\Sigma := x' \Sigma y$,

$$\left(\frac{\eta_C}{2} + \frac{\eta_U m_C}{m_U} \right) \|\theta^*\|_\Sigma^2 \leq \frac{\eta_U}{m_U} \|\theta^*\|_\Sigma \|\bar{\theta}\|_\Sigma,$$

which gives $\|\theta^*\|_\Sigma \leq \rho^*$ as in (S.3).

Step 3: Brouwer on the truncated best response. Let B be the ball of Step 0, a compact convex set with 0 in its interior. Define $\text{BR}_\rho(\theta) := \arg \max_{\theta' \in C(\theta) \cap B} V(\theta'; \theta)$ for $\theta \in B$; this is single-valued (strict concavity, nonempty compact convex feasible set) and maps B into B . The correspondence $\theta \mapsto C(\theta) \cap B$ is compact- and convex-valued and continuous: upper hemicontinuity follows from the closed graph of $\{(\theta', \theta) \in B \times B : g(\theta', \theta) \leq 0\}$ (joint continuity of g , Step 0) intersected with the compact B . For lower hemicontinuity, take $\theta_n \rightarrow \theta$ and $\theta' \in C(\theta) \cap B$, and set $\theta'_n := t_n \theta'$ with $t_n := \min\{1, W_0^C / (W_0^C + \max\{g(\theta', \theta_n), 0\})\}$. By convexity of $g(\cdot, \theta_n)$ and $g(0, \theta_n) = -W_0^C < 0$,

$$g(t_n \theta', \theta_n) \leq t_n g(\theta', \theta_n) - (1 - t_n) W_0^C \leq 0,$$

so $\theta'_n \in C(\theta_n) \cap B$ (the ball is preserved since $t_n \leq 1$), and $t_n \rightarrow 1$ because $g(\theta', \theta_n) \rightarrow g(\theta', \theta) \leq 0$.

Hence $\theta'_n \rightarrow \theta'$. Since V is jointly continuous, Berge's maximum theorem implies BR_ρ is continuous, and Brouwer's fixed-point theorem yields $\theta^* = \text{BR}_\rho(\theta^*) \in B$.

Step 4: the truncation does not bind. The fixed point θ^* maximizes $V(\cdot; \theta^*)$ over $C(\theta^*) \cap B \ni 0$, so Step 2 gives $\|\theta^*\|_\Sigma \leq \rho^* < \rho$: θ^* is interior to B . If some $\theta' \in C(\theta^*)$ had $V(\theta'; \theta^*) > V(\theta^*; \theta^*)$, then points on the segment $[\theta^*, \theta']$ near θ^* would lie in $C(\theta^*) \cap B$ (convexity of C , interiority in B) and, by concavity of V , have strictly higher value than θ^* , a contradiction. Hence $\theta^* = \text{BR}(\theta^*)$ and, by Step 1, an equilibrium exists. Finally, by Corollary 1, if (28) holds the slack CAPM allocation is an equilibrium, and if it fails every equilibrium, including the one just constructed, has $\lambda > 0$. $\square$

Proofs

Proof of Proposition 1. Fix the benchmark scale $\phi > 0$. By (2), $\Sigma - \Sigma_\varepsilon = M\Sigma_f M'$. Let P be the orthogonal projector onto $\text{col}(M_r)$ and $P_\perp = I - P$, and write $M_u^\perp := P_\perp M_u$ for the projection of the unregulated loadings onto the complement of the regulated span. Every admissible regulatory matrix $\Sigma_K = M_r \Omega M_r'$ (with $\Omega = D_w \Gamma D_w \succeq 0$) has row and column spaces contained in $\text{col}(M_r)$, so $P_\perp \Sigma_K = \Sigma_K P_\perp = 0$, and therefore $P_\perp \Delta P_\perp = -\phi^2 P_\perp M\Sigma_f M' P_\perp$. Since $P_\perp M_r = 0$, the three terms of

$$M\Sigma_f M' = M_r \Sigma_{rr} M_r' + M_r \Sigma_{ru} M_u' + M_u \Sigma_{ur} M_r' + M_u \Sigma_{uu} M_u'$$

that carry an outer M_r vanish under the compression, leaving $P_\perp \Delta P_\perp = -\phi^2 M_u^\perp \Sigma_{uu} M_u^{\perp'}$. A projector compression does not increase any unitarily invariant norm ($\|P_\perp\| = 1$), so $\|\Delta\| \geq \|P_\perp \Delta P_\perp\|$, which is (11). Its right-hand side depends on M_u and Σ_{uu} alone, not on (w, Γ) . The matrix $M_u^\perp \Sigma_{uu} M_u^{\perp'}$ is positive semidefinite, and since $\Sigma_{uu} \succ 0$ (a diagonal block of $\Sigma_f \succ 0$) it is nonzero if and only if $M_u^\perp \neq 0$, i.e. $\text{col}(M_u) \not\subseteq \text{col}(M_r)$. In the complementary case $\text{col}(M_u) \subseteq \text{col}(M_r)$ the regulated factors span all systematic risk, and the neutral calibration of Section 3.2 attains $\Sigma_K = \phi^2(\Sigma - \Sigma_\varepsilon)$, so $\Delta = 0$. $\square$

Proof of Lemma 1. (i) V^U is strictly concave ($\Sigma \succ 0$) and coercive, so the first-order condition $\mu - R_f p - \eta_U \Sigma \theta = 0$ is necessary and sufficient, giving $\theta^U = \eta_U^{-1} \Sigma^{-1} \pi$.

(ii) Write $F(\theta) := \psi \sum_i p_i |\theta_i| + \mathcal{K}(\theta)$. Since $p_i > 0$, the funding term is a weighted ℓ_1 norm and $\mathcal{K}$ is a seminorm, so F is convex, continuous, and positively homogeneous of degree one. The feasible set $C := \{\theta : F(\theta) \leq W_0^C\}$ is therefore closed and convex and contains $\theta = 0$ with strict inequality since $W_0^C > 0$, so Slater's condition holds. V^C is strictly concave and coercive, so a unique maximizer $\theta^C \in C$ exists, and by Slater there is $\lambda \geq 0$ such that θ^C maximizes the Lagrangian $V^C(\theta) - \lambda (F(\theta) - W_0^C)$ over $\mathbb{R}^N$ and complementary slackness holds. The Lagrangian is concave, and its supergradient condition is

$$0 \in \pi - \eta_C \Sigma \theta^C - \lambda \psi D_\varsigma p - \lambda \partial \mathcal{K}(\theta^C), \quad \varsigma_i \in \partial |\theta_i^C|,$$

where $\partial |\theta_i| = \{\text{sign}(\theta_i)\}$ for $\theta_i \neq 0$ and $[-1, 1]$ at $\theta_i = 0$, which is the definition of D_ς in (19).

Wherever $\mathcal{K}(\theta^C) = \kappa > 0$ the charge is differentiable with $\partial\mathcal{K}(\theta^C) = \{\Sigma_{\mathcal{K}}\theta^C/\kappa\}$ by (8), so

$$\pi - \lambda \psi D_{\varsigma} p = \left(\eta_C \Sigma + \frac{\lambda}{\kappa} \Sigma_{\mathcal{K}} \right) \theta^C = H(\lambda, \kappa) \theta^C.$$

Since $\Sigma \succ 0$, $\Sigma_{\mathcal{K}} \succeq 0$, $\eta_C > 0$, and $\lambda/\kappa \geq 0$, it follows that $H(\lambda, \kappa) \succ 0$. Inverting and using $\pi = \mu - R_f p$ gives (22). $\square$

Proof of Proposition 2. From the constrained first-order condition, $\eta_C \Sigma \theta^C = \pi - \lambda \psi D_{\varsigma} p - (\lambda/\kappa) \Sigma_{\mathcal{K}} \theta^C$, and hence

$$\theta^C = \frac{1}{\eta_C} \Sigma^{-1} \left[\pi - \lambda \psi D_{\varsigma} p - \frac{\lambda}{\kappa} \Sigma_{\mathcal{K}} \theta^C \right].$$

Substitute this and $\theta^U = \eta_U^{-1} \Sigma^{-1} \pi$ into market clearing (23):

$$\left(\frac{m_U}{\eta_U} + \frac{m_C}{\eta_C} \right) \Sigma^{-1} \pi - \frac{m_C}{\eta_C} \Sigma^{-1} \left[\lambda \psi D_{\varsigma} p + \frac{\lambda}{\kappa} \Sigma_{\mathcal{K}} \theta^C \right] = \bar{\theta}.$$

Multiplying through by $\bar{\eta} \Sigma$ and using $\chi = m_C \bar{\eta} / \eta_C$ yields (24). Since $\bar{\eta}^{-1} = m_U / \eta_U + m_C / \eta_C > m_C / \eta_C$, it follows that $\chi \in (0, 1)$. Equation (25) follows by substituting $\pi = \mu - R_f p$ into (24) and solving row by row for p_i ; the coefficient $R_f + \chi \lambda \psi \varsigma_i$ is positive because $\varsigma_i \geq -1$ and $\chi \lambda \psi < R_f$. When $\lambda = 0$ the two distortion terms vanish. $\square$

Proof of Proposition 3. Divide row i of (24) by $p_i > 0$ and use $\mathbb{E}[R_i] - R_f = \pi_i / p_i$. For the fundamental term,

$$\frac{\bar{\eta} (\Sigma \bar{\theta})_i}{p_i} = \bar{\eta} p_M \frac{\text{Cov}(X_i, \bar{\theta}' X)}{p_i p_M} = \bar{\eta} p_M \text{Cov}(R_i, R_M) = \beta_i^M \cdot \bar{\eta} p_M \text{Var}(R_M) = \beta_i^M \pi_M.$$

For the funding term, $\chi \lambda \psi \varsigma_i p_i / p_i = \psi \chi \lambda \varsigma_i$. For the regulatory term, by the definitions of $\text{Cov}_{\mathcal{K}}$ and $\text{Var}_{\mathcal{K}}$,

$$\frac{\chi \lambda}{\kappa} \frac{(\Sigma_{\mathcal{K}} \theta^C)_i}{p_i} = \frac{\chi \lambda}{\kappa} p_C \text{Cov}_{\mathcal{K}}(R_i, R_C) = \beta_i^{\mathcal{K}} \cdot \frac{\chi \lambda}{\kappa} p_C \text{Var}_{\mathcal{K}}(R_C) = \beta_i^{\mathcal{K}} \frac{\chi \lambda \kappa}{p_C},$$

where the last equality uses $\text{Var}_{\mathcal{K}}(R_C) = \theta^{C'} \Sigma_{\mathcal{K}} \theta^C / p_C^2 = \kappa^2 / p_C^2$. $\square$

Proof of Corollary 1. In any equilibrium with $\lambda = 0$, both sectors' demands are proportional to $\Sigma^{-1} \pi$, so clearing pins down $\pi = \bar{\eta} \Sigma \bar{\theta}$, prices $p^0 = R_f^{-1}(\mu - \bar{\eta} \Sigma \bar{\theta})$, and $\theta^C = (\bar{\eta} / \eta_C) \bar{\theta}$. This candidate is an equilibrium if and only if θ^C is feasible at $\lambda = 0$, i.e. $\psi \sum_i p_i^0 |\theta_i^C| + \mathcal{K}(\theta^C) \leq W_0^C$. Since $\bar{\theta} \in \mathbb{R}_+^N$ the frictionless book is long in every asset, so the funding term equals $\psi p^{0'} \theta^C = \psi (\bar{\eta} / \eta_C) p^{0'} \bar{\theta}$. By homogeneity of degree one of $\mathcal{K}$, $\mathcal{K}((\bar{\eta} / \eta_C) \bar{\theta}) = (\bar{\eta} / \eta_C) \mathcal{K}(\bar{\theta})$, which gives (28). Conversely, if (28) fails, no equilibrium can have $\lambda = 0$ (the unique $\lambda = 0$ candidate is infeasible), so every equilibrium has $\lambda > 0$. $\square$

The next result, referenced in Section 4.3 of the main text, states the Black-type benchmark in closed form.

Corollary S.1 (A covering, neutrally calibrated rule raises returns in proportion to size). *Suppose $\Sigma_K = \phi^2 \Sigma$ for some $\phi > 0$. In any binding equilibrium ($\lambda > 0$, $\kappa = \mathcal{K}(\theta^C) > 0$), expected returns obey the exact one-factor relation*

$$\mathbb{E}[R_i] - R_f = b \psi \varsigma_i + a p_M \text{Cov}(R_i, R_M), \quad i = 1, \dots, N,$$

where, with $\tilde{a} := \eta_C + \lambda \phi^2 / \kappa$ and $\varpi := \chi \lambda \phi^2 / (\kappa \tilde{a}) \in (0, 1)$,

$$a = \frac{\bar{\eta}}{1 - \varpi} > \bar{\eta}, \quad b = \frac{\lambda \chi \eta_C}{\tilde{a} (1 - \varpi)} > 0.$$

The requirement steepens the security market line and, under a self-financed book, raises the zero-beta intercept on every asset the constrained sector holds long, but it leaves relative valuations undistorted.

Proof of Corollary S.1. With $\Sigma_K = \phi^2 \Sigma$ and $\kappa > 0$, $H(\lambda, \kappa) = \tilde{a} \Sigma$ with $\tilde{a} := \eta_C + \lambda \phi^2 / \kappa$, so (22) gives $\Sigma \theta^C = (\pi - \lambda \psi D_\varsigma p) / \tilde{a}$. Substituting into (24),

$$\pi = \bar{\eta} \Sigma \bar{\theta} + \chi \lambda \psi D_\varsigma p + \varpi (\pi - \lambda \psi D_\varsigma p), \quad \varpi := \frac{\chi \lambda \phi^2}{\kappa \tilde{a}}.$$

Since $\chi < 1$ and $\lambda \phi^2 / \kappa < \tilde{a}$, it follows that $\varpi \in (0, 1)$. Collecting terms,

$$\pi = a \Sigma \bar{\theta} + b \psi D_\varsigma p, \quad a := \frac{\bar{\eta}}{1 - \varpi} > \bar{\eta}, \quad b := \frac{\lambda (\chi - \varpi)}{1 - \varpi} = \frac{\lambda \chi \eta_C}{\tilde{a} (1 - \varpi)} > 0,$$

using $\chi - \varpi = \chi (1 - \lambda \phi^2 / (\kappa \tilde{a})) = \chi \eta_C / \tilde{a}$. Dividing row i by p_i and using the identity $(\Sigma \bar{\theta})_i / p_i = p_M \text{Cov}(R_i, R_M)$ gives the return-space statement. $\square$

Proof of Corollary 2. $M_r' z = 0$ implies $\Sigma_K z = M_r \Omega M_r' z = 0$, and by symmetry $z' \Sigma_K \theta^C = (\Sigma_K z)' \theta^C = 0$, so $\beta_z^K = 0$. Premultiplying (24) by z' gives

$$z' \pi = \bar{\eta} z' \Sigma \bar{\theta} + \chi \lambda \psi z' D_\varsigma p = \bar{\eta} z' \Sigma \bar{\theta} + \chi \lambda \psi \varsigma_z p' z.$$

Dividing by $p' z \neq 0$ and writing the result in return form yields the claim. $\square$

Proof of Proposition 4. Fix $(\lambda, \kappa) \in [0, \infty) \times (0, \infty)$ and a configuration ς , write $H = H(\lambda, \kappa) \succ 0$ and $Z := \{i : \varsigma_i = 0\}$. In any candidate price system the traditional investors' first-order condition and market clearing give (S.2), so that

$$\pi(\theta^C) = \frac{\eta_U}{m_U} \Sigma (\bar{\theta} - m_C \theta^C), \quad p(\theta^C) = \hat{p} + c \Sigma \theta^C, \quad \hat{p} := \frac{1}{R_f} \left[\mu - \frac{\eta_U}{m_U} \Sigma \bar{\theta} \right], \quad c := \frac{\eta_U m_C}{m_U R_f},$$

are affine in the bank's book, and $\theta^U = (\bar{\theta} - m_C \theta^C) / m_U$. Substituting into the constrained first-order condition $H \theta^C = \pi - \lambda \psi D_\varsigma p$ gives, for the rows $i \notin Z$ and with $\theta_i^C = 0$ for $i \in Z$,

$$\left[\frac{\eta_U m_C}{m_U} \Sigma + \lambda \psi c D_\varsigma \Sigma + H \right] \theta^C = \frac{\eta_U}{m_U} \Sigma \bar{\theta} - \lambda \psi D_\varsigma \hat{p},$$

while the rows $i \in Z$ of the first-order condition determine the free products $u_i := \varsigma_i p_i = [\pi(\theta^C) - H \theta^C]_i / (\lambda \psi)$ and hence $\varsigma_i = u_i / p_i(\theta^C)$. The system is therefore linear in (θ_{-Z}^C, u_Z) , with (p, θ^U)

affine in the solution, and it is uniquely solvable if and only if the principal submatrix $M_\varsigma(\lambda, \kappa)$ of $M := (\eta_U m_C / m_U) \Sigma + \lambda \psi c D_\varsigma \Sigma + H$ on the rows and columns outside Z is nonsingular. When $\varsigma_i = 1$ for every $i \notin Z$, or when $\psi = 0$, $M = (\eta_U m_C / m_U + \lambda \psi c) \Sigma + H \succ 0$, every principal submatrix is nonsingular, and the closed form is

$$\theta^C(\lambda, \kappa) = M^{-1} \left[\frac{\eta_U}{m_U} \Sigma \bar{\theta} - \lambda \psi \hat{p} \right], \quad p = \hat{p} + c \Sigma \theta^C, \quad \theta^U = \frac{\bar{\theta} - m_C \theta^C}{m_U}$$

(for $Z = \emptyset$; otherwise the same expressions on the complement of Z). In general $\det M_\varsigma(\lambda, \kappa)$ is a polynomial in $(\lambda, \lambda/\kappa)$ that equals the corresponding principal minor of $(\eta_U m_C / m_U + \eta_C) \Sigma \succ 0$ at $\lambda = 0$, so it is not identically zero, and its zero set is a proper algebraic subset of $[0, \infty) \times (0, \infty)$, closed and of measure zero. Given this map, a binding-constraint equilibrium requires that (a) the conjectured κ equal the realized charge, $\mathcal{K}(\theta^C(\lambda, \kappa)) = \kappa$, so that (22) is the true optimality condition, (b) the constraint hold with equality, $\psi \sum_i p_i |\theta_i^C| + \kappa = W_0^C$, consistent with $\lambda > 0$ and complementary slackness, and (c) the configuration be self-consistent, so that D_ς is a selection from the subdifferential of the funding term at θ^C . Conversely any $(\lambda, \kappa, \varsigma)$ satisfying (a)–(c) generates portfolios and prices satisfying optimality (Lemma 1) and market clearing, hence an equilibrium. $\square$

Proof of Proposition 5. Write $F(\theta) := \psi \sum_i p_i |\theta_i| + \mathcal{K}(\theta)$, which is convex since $p_i > 0$. Monotonicity is immediate since the feasible set $C(W) := \{\theta : F(\theta) \leq W\}$ expands with W . For concavity, let θ_i be optimal at W_i ($i = 1, 2$) and $t \in [0, 1]$. By convexity of F , the portfolio $t\theta_1 + (1-t)\theta_2$ is feasible at $tW_1 + (1-t)W_2$, and concavity of V^C gives $v(tW_1 + (1-t)W_2) \geq t v(W_1) + (1-t) v(W_2)$.

For the supergradient property, fix $W > 0$ with optimum $\theta(W)$ and multiplier $\lambda(W)$. Slater's condition ($\theta = 0$) implies the saddle-point property: for all $\theta' \in \mathbb{R}^N$,

$$V^C(\theta') - \lambda(W) [F(\theta') - W] \leq V^C(\theta(W)) - \lambda(W) [F(\theta(W)) - W] = v(W),$$

using complementary slackness for the equality. If θ' is feasible at W' , then $F(\theta') - W \leq W' - W$, so $V^C(\theta') \leq v(W) + \lambda(W)(W' - W)$. Taking the supremum over θ' feasible at W' gives $v(W') \leq v(W) + \lambda(W)(W' - W)$.

Monotonicity of multipliers follows: for $W < W'$, the supergradient inequalities at W and W' give

$$\lambda(W')(W' - W) \leq v(W') - v(W) \leq \lambda(W)(W' - W),$$

hence $\lambda(W') \leq \lambda(W)$. A concave function is differentiable almost everywhere, and at points of differentiability the supergradient is unique, so $v'(W) = \lambda(W)$ there. $\square$

Proof of Proposition 6. Throughout, “premium” means $\mu - R_f p$. Note $W < \bar{W}$ requires $\bar{W} > 0$, hence $\psi p^0 + w > 0$.

Step 0: any equilibrium has $\theta^C > 0$, $\psi p + w > 0$, and a positive premium. In a slack equilibrium ($\lambda = 0$), Corollary 1 pins the CAPM allocation, which is feasible if and only if $W \geq \bar{W}$. Since $W < \bar{W}$, every equilibrium is binding, $\lambda > 0$, with the constraint holding with equality: $(\psi p +$

$w)|\theta^C| = W > 0$. This rules out $\theta^C = 0$ and gives $\psi p + w > 0$. If $\theta^C < 0$, clearing gives $\theta^U = (\bar{\theta} - m_C \theta^C)/m_U > \bar{\theta}/m_U > 0$, so the premium $\eta_U \sigma^2 \theta^U$ is strictly positive, and the bank's first-order condition $\mu - R_f p - \eta_C \sigma^2 \theta^C = -\lambda(\psi p + w)$, the right side being λ times the derivative of $\theta \mapsto (\psi p + w)|\theta|$ at $\theta^C < 0$, has a strictly positive left side and a strictly negative right side, a contradiction. Hence $\theta^C > 0$, the binding constraint reads $(\psi p + w)\theta^C = W$, and the bank's first-order condition $\mu - R_f p = \eta_C \sigma^2 \theta^C + \lambda(\psi p + w) > 0$ gives a positive premium and $\theta^U > 0$.

Step 1: uniqueness and (32). By Step 0, any equilibrium satisfies $\theta^C = W/(\psi p + w)$ and $\theta^U = (\mu - R_f p)/(\eta_U \sigma^2)$, so clearing is (31). On $(-w/\psi, \infty)$ the left side of (31) is strictly decreasing (both terms are), tends to $+\infty$ as $p \downarrow -w/\psi$, and to $-\infty$ as $p \rightarrow \infty$; at $\psi = 0$ the second term is constant, the first is strictly decreasing on $\mathbb{R}$ with limits $\pm\infty$, and the same conclusion holds. Hence there is a unique root p^* . Substituting clearing into the bank's first-order condition,

$$\lambda(\psi p^* + w) = \mu - R_f p^* - \eta_C \sigma^2 \theta^C = \frac{\eta_U \sigma^2}{m_U} (\bar{\theta} - m_C \theta^C) - \eta_C \sigma^2 \theta^C = \sigma^2 \left[\frac{\eta_U}{m_U} \bar{\theta} - \left(\frac{\eta_U m_C}{m_U} + \eta_C \right) \theta^C \right],$$

which is (32). Since $(\eta_U/m_U)/(\eta_U m_C/m_U + \eta_C) = \bar{\eta}/\eta_C$, it follows that $\lambda > 0$ if and only if $\theta^C < (\bar{\eta}/\eta_C)\bar{\theta}$. To verify this inequality, note that at $W = \bar{W}$ the pair $(p, \theta^C) = (p^0, (\bar{\eta}/\eta_C)\bar{\theta})$ solves (31) (direct substitution, using $\mu - R_f p^0 = \bar{\eta} \sigma^2 \bar{\theta}$ and $1/\bar{\eta} = m_U/\eta_U + m_C/\eta_C$). By Step 2 below, θ^C is strictly increasing in W . Since $W < \bar{W}$, it follows that $\theta^C < (\bar{\eta}/\eta_C)\bar{\theta}$ and $\lambda > 0$. The candidate then satisfies the sufficient KKT conditions of the concave bank problem, so it is an equilibrium, and by the uniqueness of p^* it is the only one.

Step 2: comparative statics in W . Raising W raises the left side of (31) pointwise on $(-w/\psi, \infty)$. Since that function is strictly decreasing in p , the root p^* strictly increases. Hence $\theta^U = (\mu - R_f p^*)/(\eta_U \sigma^2)$ and the premium strictly decrease, and $\theta^C = (\bar{\theta} - m_U \theta^U)/m_C$ strictly increases. In (32), the bracket is positive and strictly decreasing in W (via θ^C), and the factor $1/(\psi p^* + w)$ is nonincreasing in W (via p^* ; constant at $\psi = 0$). Hence λ is strictly decreasing.

Step 3: comparative statics in w . At fixed p , raising w lowers the left side of (31), so p^* strictly decreases, θ^U and the premium strictly increase, and θ^C strictly decreases (clearing). For the limit, Step 0 bounds the premium by $\eta_U \sigma^2 \bar{\theta}/m_U$ from above (since $\theta^C > 0$ implies $\theta^U < \bar{\theta}/m_U$), so p^* is bounded below and $\psi p^* + w \rightarrow \infty$ as $w \rightarrow \infty$. Then $\theta^C = W/(\psi p^* + w) \rightarrow 0$, $\theta^U \rightarrow \bar{\theta}/m_U$, and the premium converges to $\eta_U \sigma^2 \bar{\theta}/m_U$. From (32), $0 < \lambda \leq \sigma^2 (\eta_U/m_U) \bar{\theta}/(\psi p^* + w) \rightarrow 0$. Finally, $\bar{W}$ is strictly increasing in w , so the binding region is $\{w > \bar{w}\}$ with $\bar{W} = W$ at $w = \bar{w}$. The root $p^*(w)$ varies continuously (a strictly monotone, continuous family of clearing equations), and at $w = \bar{w}$ Step 1 gives $\theta^C = (\bar{\eta}/\eta_C)\bar{\theta}$, so $\lambda \rightarrow 0$ as $w \downarrow \bar{w}$. Since $\lambda > 0$ on $(\bar{w}, \infty)$ and $\lambda \rightarrow 0$ at both ends, λ attains an interior maximum and is nonmonotone, while by the first part of this step the premium is strictly increasing throughout. $\square$

Proof of Proposition 7. The traditional first-order condition is $\eta_U \Sigma \theta^U = \pi$. The bank's (Lemma 1, with $\kappa > 0$) is $\eta_C \Sigma \theta^C = \pi - \lambda \psi D_\psi p - (\lambda/\kappa) \Sigma_\kappa \theta^C$. Substituting the former into the latter and

multiplying by Σ^{-1} ,

$$\eta_C \theta^C = \eta_U \theta^U - \lambda \tau, \quad \tau = \Sigma^{-1} [\psi D_\zeta p + \Sigma_K \theta^C / \kappa] = \Sigma^{-1} [\psi D_\zeta p + \nabla \mathcal{K}(\theta^C)].$$

Substituting $\theta^U = (\bar{\theta} - m_C \theta^C) / m_U$ from market clearing and collecting terms,

$$\left(\eta_C + \frac{\eta_U m_C}{m_U} \right) \theta^C = \frac{\eta_U}{m_U} \bar{\theta} - \lambda \tau.$$

Since $\eta_C + \eta_U m_C / m_U = (\eta_C m_U + \eta_U m_C) / m_U = \eta_U \eta_C / (\bar{\eta} m_U)$, this gives $\theta^C = (\bar{\eta} / \eta_C) \bar{\theta} - (\bar{\eta} m_U \lambda / (\eta_U \eta_C)) \tau$, and $\bar{\eta} m_U / \eta_U = 1 - \chi$ yields the first equation in (33). Then, from clearing,

$$\theta^U = \frac{\bar{\theta} - m_C \theta^C}{m_U} = \frac{1}{m_U} \left(1 - \frac{m_C \bar{\eta}}{\eta_C} \right) \bar{\theta} + \frac{m_C (1 - \chi) \lambda}{m_U \eta_C} \tau = \frac{\bar{\eta}}{\eta_U} \bar{\theta} + \frac{\chi \lambda}{\eta_U} \tau,$$

using $1 - m_C \bar{\eta} / \eta_C = \bar{\eta} m_U / \eta_U$ and $m_C (1 - \chi) / (m_U \eta_C) = m_C \bar{\eta} / (\eta_U \eta_C) = \chi / \eta_U$. The deviations aggregate to zero because $m_C (1 - \chi) / \eta_C = m_U \chi / \eta_U (= m_U m_C \bar{\eta} / (\eta_U \eta_C))$. The decomposition of τ is immediate from $\nabla \mathcal{K}(\theta^C) = M_r \Omega s(\theta^C) / \kappa$, whose Σ^{-1} -image lies in $\Sigma^{-1} \text{col}(M_r)$. $\square$

Proof of Corollary 3. Because a and b are uncorrelated with each other and with every other asset under the true covariance Σ , the a - and b -rows of $\Sigma \theta$ read $(\Sigma \theta)_a = \sigma^2 \theta_a$ and $(\Sigma \theta)_b = \sigma^2 \theta_b$ for any book θ . On the regulatory side, b loads only on an unregulated factor, so its row of M_r vanishes and $(\Sigma_K \theta)_b = 0$; asset a loads on a single regulated factor of risk weight w that is Γ -uncorrelated with every other regulated exposure, so $(\Sigma_K \theta)_a = w^2 \theta_a$. The a - and b -rows of both sectors' first-order conditions therefore decouple from the rest of the book. Market clearing for each twin is $m_U \theta_i^U + m_C \theta_i^C = \bar{\theta}_0$, and the traditional condition gives $\theta_i^U = \pi_i / (\eta_U \sigma^2)$.

Signs. Since $\theta_b^C > 0$ by hypothesis, $\varsigma_b = 1$ and the b -row of (19) reads $\pi_b - \lambda \psi p_b = \eta_C \sigma^2 \theta_b^C > 0$. Suppose, towards a contradiction, that $\theta_a^C \geq \theta_b^C$. Clearing then gives $\theta_a^U \leq \theta_b^U$, hence $\pi_a \leq \pi_b$ and $p_a \geq p_b$. But the a -row of (19) with $\theta_a^C > 0$ (so $\varsigma_a = 1$) reads $\pi_a - \lambda \psi p_a = \eta_C \sigma^2 (1 + \zeta) \theta_a^C$ with $\zeta := \lambda w^2 / (\kappa \eta_C \sigma^2) > 0$, so

$$\eta_C \sigma^2 (1 + \zeta) \theta_a^C = \pi_a - \lambda \psi p_a \leq \pi_b - \lambda \psi p_b = \eta_C \sigma^2 \theta_b^C,$$

which forces $\theta_a^C \leq \theta_b^C / (1 + \zeta) < \theta_b^C$, a contradiction. Hence $\theta_a^C < \theta_b^C$, and clearing gives $\theta_a^U > \theta_b^U$, $\pi_a > \pi_b$, and $p_a < p_b$. (When $\theta_a^C \leq 0$ the conclusion $\theta_a^C < \theta_b^C$ is immediate and the same clearing argument applies.)

Closed form. Suppose in addition $\theta_a^C > 0$, so that $\varsigma_a = \varsigma_b = 1$. Writing $\tilde{\pi}_i := \pi_i - \lambda \psi p_i = (1 + \lambda \psi / R_f) \pi_i - (\lambda \psi / R_f) \mu_0$ (using $p_i = (\mu_0 - \pi_i) / R_f$), the two bank rows give

$$\theta_a^C = \frac{\tilde{\pi}_a}{\eta_C \sigma^2 (1 + \zeta)}, \quad \theta_b^C = \frac{\tilde{\pi}_b}{\eta_C \sigma^2}.$$

Subtracting the two clearing equations and using $\tilde{\pi}_a - \tilde{\pi}_b = (1 + \lambda \psi / R_f) (\pi_a - \pi_b)$,

$$\left[\frac{m_U}{\eta_U} + \frac{m_C}{\eta_C} \frac{1 + \lambda \psi / R_f}{1 + \zeta} \right] \frac{\pi_a - \pi_b}{\sigma^2} = \frac{m_C}{\eta_C \sigma^2} \tilde{\pi}_b \frac{\zeta}{1 + \zeta}.$$

Since $\tilde{\pi}_b = \eta_C \sigma^2 \theta_b^C > 0$, this gives (35). For the comparative statics, write (35) as $F(\zeta) =$

$c_1 \zeta / [c_2(1 + \zeta) + c_3]$ with $c_1 = m_C \sigma^2 \theta_b^C$, $c_2 = m_U / \eta_U$, $c_3 = (m_C / \eta_C)(1 + \lambda \psi / R_f)$. Then $F'(\zeta) = c_1(c_2 + c_3) / [c_2(1 + \zeta) + c_3]^2 > 0$, ζ is strictly increasing in w , and $F(\zeta) \rightarrow c_1/c_2 = \eta_U m_C \sigma^2 \theta_b^C / m_U$ as $\zeta \rightarrow \infty$. $\square$

Proof of Proposition 9. For any θ , completing the square gives

$$V^j(\theta) = \frac{1}{2\eta_j} \pi' \Sigma^{-1} \pi - \frac{\eta_j}{2} \left\| \theta - \frac{1}{\eta_j} \Sigma^{-1} \pi \right\|_{\Sigma}^2.$$

For the traditional investor the second term is zero at the optimum, so $V^U = \frac{1}{2\eta_U} \pi' \Sigma^{-1} \pi$ in either equilibrium. Proposition 2 gives $\pi = \pi^0 + \chi \lambda \Sigma \tau$, and it follows that

$$\pi' \Sigma^{-1} \pi - \pi^{0'} \Sigma^{-1} \pi^0 = 2\chi \lambda \pi^{0'} \tau + (\chi \lambda)^2 \|\tau\|_{\Sigma}^2 = 2\chi \lambda \bar{\eta} T + (\chi \lambda)^2 \|\tau\|_{\Sigma}^2,$$

using $\pi^0 = \bar{\eta} \Sigma \bar{\theta}$ and $\Sigma \tau = \psi D_{\zeta} p + \nabla \mathcal{K}(\theta^C)$. Dividing by $2\eta_U$ gives (i), and $T \geq 0$ makes both terms nonnegative. For (ii), the first-order condition (19) rearranges to $\frac{1}{\eta_C} \Sigma^{-1} \pi - \theta^C = \frac{\lambda}{\eta_C} \tau$, so the square-completion term equals $\frac{\lambda^2}{2\eta_C} \|\tau\|_{\Sigma}^2$, the compliance cost. The first term changes by the premium difference above scaled by $\frac{1}{2\eta_C}$, the scarcity gain. For (iii), $R_f(p^0 - p)' \bar{\theta} = (\pi - \pi^0)' \bar{\theta} = \chi \lambda \tau' \Sigma \bar{\theta} = \chi \lambda T$. For (iv), form $m_U \Delta C E^U + m_C \Delta C E^C - \chi \lambda T$. The terms linear in T collect to $\chi \lambda T [\bar{\eta} (m_U / \eta_U + m_C / \eta_C) - 1] = 0$ by the definition of $\bar{\eta}$. The quadratic terms collect to

$$\frac{\lambda^2 \|\tau\|_{\Sigma}^2}{2} \left[\chi^2 \left(\frac{m_U}{\eta_U} + \frac{m_C}{\eta_C} \right) - \frac{m_C}{\eta_C} \right] = \frac{\lambda^2 \|\tau\|_{\Sigma}^2}{2} \cdot \frac{\chi^2 - \chi}{\bar{\eta}} = -\frac{\chi(1 - \chi)}{2\bar{\eta}} \lambda^2 \|\tau\|_{\Sigma}^2,$$

using $m_C / \eta_C = \chi / \bar{\eta}$. Since $\chi \in (0, 1)$, the sum is $-DWL \leq 0$. Equivalently, expected aggregate payoffs are invariant to the allocation of the fixed supply, so the change in total surplus equals the change in aggregate variance penalties, and those penalties are minimized by the proportional-sharing allocation $\theta^j = (\bar{\eta} / \eta_j) \bar{\theta}$, which is the frictionless equilibrium allocation. $\square$

Assembling the sensitivity gap from supervisory data

This section, referenced in Section 5.4 of the main text, shows how the sensitivity gap of Definition 2 is built from reported data.

Both ingredients of (37) are computable from data the regime already generates, so the gap is built to be computed rather than estimated. Factor by factor,

$$g_k = w_k (\chi s_k^{\text{mkt}} - s_k^{\text{bank}}), \quad (\text{S.4})$$

the difference, at regulated factor k , between the weighted sensitivity the banking sector would report in the frictionless benchmark and the weighted sensitivity it actually reports.

The bank term $w_k s_k^{\text{bank}} = \sum_b \text{WS}_k^b$ is the sum, across standardized-approach banks b , of the weighted sensitivities $\text{WS}_k = w_k s_k$ that are line items of the SBM aggregation itself (MAR21.4). Every reporting bank already computes them to produce its own charge, so a supervisor aggregating them by factor across banks is feasible.

The benchmark term $\chi w_k s_k^{\text{mkt}}$ needs the regulated-factor weighted sensitivities of the outstand-

Table S.1: Illustrative sensitivity-gap dashboard, reconstructed from the calibrated economy of Section 6. The benchmark column is χ times each factor’s market weighted sensitivity (capital intensity $\times$ supply, Table S.3). The bank column applies the equilibrium bank shares of Table 1. The gap is their difference (Definition 2). Entries are in the standard’s weighted-sensitivity (capital) units and are rounded independently, so the columns need not combine exactly. The norm of the gap column reproduces the baseline $G = 0.87$ of Figure 2 (self-financed book). Populated with reported supervisory weighted sensitivities, the same three columns are the operational indicator.

Regulated factor	Bank $\sum_b WS_k^b$	Benchmark χWS_k^{mkt}	Gap g_k
GIRR 10y (duration)	0.48	1.32	0.85
Equity bucket 12 (index)	1.29	1.50	0.21
Equity bucket 5 (large-cap)	0.74	0.80	0.06
Norm $G = \ g\ $			0.87

ing market portfolio, recoverable from issuance and holdings statistics through the same factor definitions, together with an estimate of the intermediary share χ . Neither needs to be exact, since both move slowly, so the informative variation in g at supervisory frequencies comes from the reported bank sensitivities themselves.

The reporting frequency is not a constraint. The Basel III monitoring exercise already collects granular market-risk data on a semiannual cycle (Basel Committee on Banking Supervision 2025), and the factor-level aggregation the gap requires is an incremental extension of it. At national supervisory-reporting frequency the bank term refreshes automatically.

Table S.1 carries out the assembly at the calibrated economy of Section 6 of the main text. Each row is a regulated factor, and the three numeric columns are what a supervisor populates.

Read across a row, the gap shows where the requirement’s effect on prices falls. At the baseline calibration the duration factor carries most of the gap (the banking sector’s reported weighted sensitivity to it is about 64% below its benchmark share, having shed duration to 12% of supply against a $\chi = 1/3$ benchmark), while the two equity factors sit closer to benchmark. Under the proportional increase in risk weights of Figure 2 the equity gaps open as banks are priced out of regulated risk, and the norm $G = \|g\|$ climbs from 0.87 to 1.77 through the region in which the shadow cost peaks and falls. The components g_k show where in the cross-section the repricing of Proposition 3 is concentrated, and none of this requires the pricing distortion to be observed, since it is read directly off the sensitivities banks file.

Supplementary figures and tables

Figure S.1, referenced in Section 6.5 of the main text, reports the effect of varying bank capital at FRTB parameters. Figures S.2 and S.3 repeat the risk-weight scaling of Figures 1 and 2 under a pure capital requirement. Table S.2, referenced in Section 5.3 of the main text, reports back-of-the-envelope marginal capital intensities and the implied pricing hits. Table S.3, referenced in Section 6.1 of the main text, reports the calibrated parameters of the numerical economy.

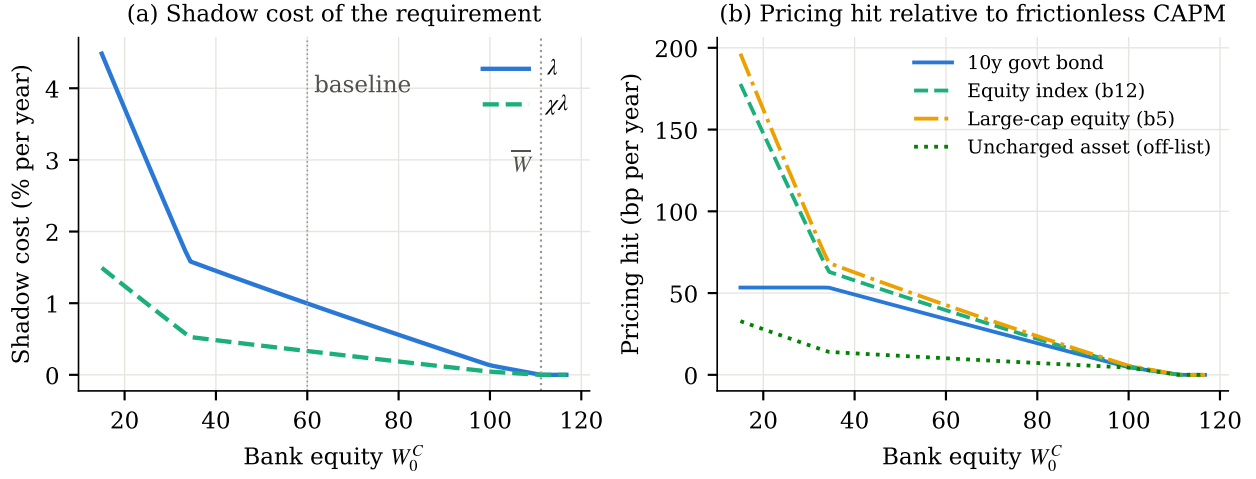


Figure S.1: The effect of varying bank capital under a self-financed book ($\psi = 1$). Panel (a): shadow cost λ and tightness statistic $\chi\lambda$. The requirement is slack beyond $\bar{W} = 111.2$. Panel (b): pricing hits by asset. All identities verified at every grid point.

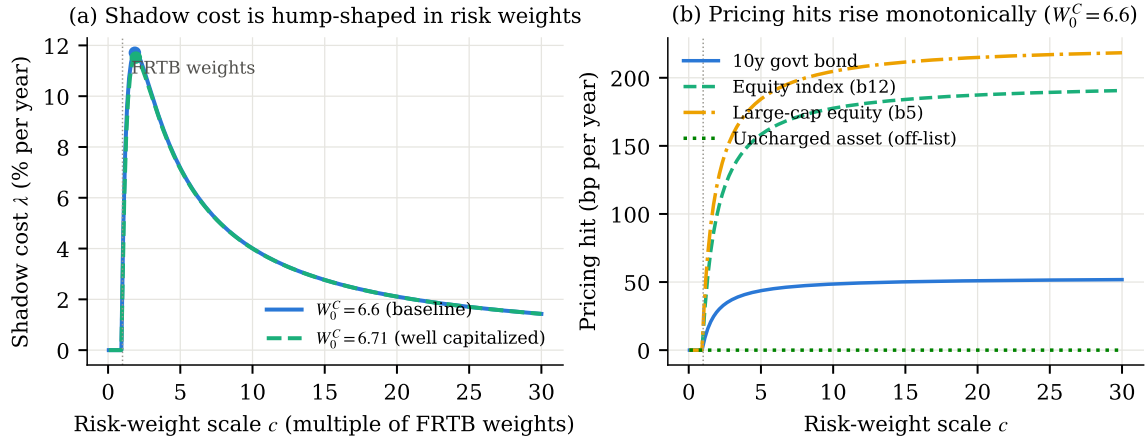


Figure S.2: A proportional change in all risk weights under a pure capital requirement ($\psi = 0$), the counterpart of Figure 1. Baseline equity is $W_0^C = 6.6$ and the well-capitalized case is $W_0^C = 6.7$, just below the threshold $\bar{W} = 7.2$. Panel (a): the shadow cost peaks near $c \approx 2$ at both equity levels. Panel (b): pricing hits at baseline equity rise monotonically throughout, and the uncharged asset bears none.

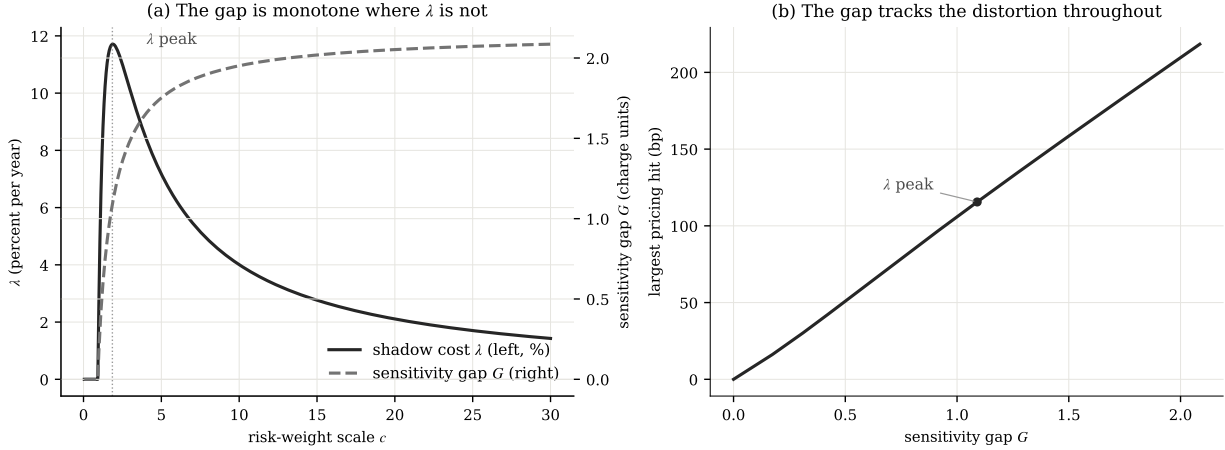


Figure S.3: The sensitivity gap under the proportional change in risk weights, pure capital requirement ($\psi = 0$), the counterpart of Figure 2. Panel (a): the shadow cost λ (solid, left axis) peaks near $c \approx 2$ and falls thereafter; the sensitivity gap G (dashed, right axis) rises monotonically from 0.21 to 2.09. Panel (b): the largest pricing hit is strictly increasing in G .

Table S.2: Marginal capital intensity and implied pricing hit (annual expected excess return relative to a zero-charge twin), by FRTB delta treatment. *Notes:* Risk weights as encoded in the project's FRTB parameter file. GIRR tenor weights are 1.7% (0.25y) declining to 1.1% (5y–30y), from MAR21.42, with the MAR21.44 $\sqrt{2}$ divisor for specified currencies. Equity bucket weights follow MAR21.77, where buckets 5 and 1 are single-name buckets and bucket 12 an index bucket, and FX is 15% (MAR21.86) with the $\sqrt{2}$ divisor for liquid pairs (MAR21.87). CSR non-securitisation parameters follow MAR21.53–21.54: the corporate row uses bucket 4 (investment-grade industrials, risk weight 3.0%) with spread duration 4.5, and the bond–CDS row uses the 99.90% basis correlation between different curves of the same issuer. Durations: 1.9 (2y), 4.5 (5y), 8.5 (10y), 17.5 (30y). Commodity rows are omitted. All figures are medium-scenario, delta-only, and ignore the MAR21.6 scenario-max, which adds a further loading that depends on book composition. They are therefore conservative.

Exposure (per \$1 market value)	FRTB treatment	Capital intensity	Pricing hit (bp) at $\chi\lambda =$		
			1%	3%	5%
2y government note, liquid ccy	GIRR 2y, RW 1.3%/√2	1.7%	2	5	9
10y government bond, liquid ccy	GIRR 10y, RW 1.1%/√2	6.6%	7	20	33
30y government bond, liquid ccy	GIRR 30y, RW 1.1%/√2	13.6%	14	41	68
5y IG corporate bond	GIRR 5y + CSR b4, RW 3.0%	17.0%	17	51	85
FX position, liquid pair	FX, RW 15%/√2	10.6%	11	32	53
Advanced-economy equity index	Equity b12, RW 15%	15.0%	15	45	75
Large-cap advanced equity	Equity b5, RW 30%	30.0%	30	90	150
Large-cap emerging equity	Equity b1, RW 55%	55.0%	55	165	275
Cross-curve 10y basis, same ccy	GIRR, $\rho = 99.90\%$	0.30%	0.3	0.9	1.5
Bond–CDS basis, 5y IG corporate	CSR, $\rho_{\text{basis}} = 99.90\%$	0.60%	0.6	1.8	3.0
On-/off-the-run basis	Same risk factor	0	0	0	0

Table S.3: Calibrated economy of Section 6 of the main text. Factor correlations: $\text{corr}(f_2, f_3) = 0.85$, $\text{corr}(f_2, f_4) = 0.50$, $\text{corr}(f_3, f_4) = 0.45$. Prescribed Γ : $\gamma_{23} = 0.45$ (equity cross-bucket), cross-class 0. $\eta_U = \eta_C = 0.05$, $m_C = 1/3$, $R_f = 1.04$. Self-financed book ($\psi = 1$): $\bar{W} = 111.2$, baseline $W_0^C = 60$. Pure capital requirement ($\psi = 0$): $\bar{W} = 7.2$, baseline $W_0^C = 6.6$.

Asset	Factor	Sys. vol	Idio. vol	Supply	FRTB delta treatment	$\mathcal{K}$ per \$1
10y govt bond	f_1 (GIRR 10y)	5.9%	0.5%	60	RW $1.1\%/\sqrt{2}$, $D = 8.5$	6.61%
Equity index	f_2 (equity b12)	16%	1%	30	RW 15%	15%
Large-cap equity	f_3 (equity b5)	19%	6%	8	RW 30%	30%
Basis package	f_4 (off-list)	5.5%	1%	6	none	0

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used Claude (Anthropic) in order to search the related literature, check derivations and proofs, draft and edit the replication code, and edit the prose for clarity. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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